Toric Geometry: A Working Reference

From fans and Cox quotients to polytopes and mirror symmetry

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Abstract

These notes are designed as both an introduction and a working reference. Each major construction is presented in the order *mental picture* → *executable recipe* → *example* → *precise statement*. The central theme is that toric geometry translates geometry into integral linear algebra: cones specify affine charts, fans specify how the charts glue, and lattice polytopes specify projective embeddings and line bundles.

The main exposition follows Chapter 7 of *Mirror Symmetry* [1], with corrections and supplementary material from *Toric Varieties* [2], Fulton's *Introduction to Toric Varieties* [4], and Cox's homogeneous-coordinate construction [5]. Basic algebraic geometry is assumed; [3] is a standard reference.

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How to use this note

Quick lookup: the 90-second reset

When a toric concept has been forgotten, first recover the following chain:

$$\boxed{\text{lattice directions } N} \longleftrightarrow \boxed{\text{monomials } M = N^\vee},$$
$$\boxed{\text{cones } \sigma} \longrightarrow \boxed{\text{affine charts } U_\sigma} \longrightarrow \boxed{\text{fan } \Sigma \text{ and } X_\Sigma}.$$

A ray gives a divisor, a maximal cone gives a torus fixed point, and inclusion of cones reverses inclusion of orbit closures. A polygon lives in the dual lattice M; its primitive inward edge normals live in N and form its normal fan. Thus the fastest route from a polygon to a variety is

$$\boxed{\text{polygon}} \xrightarrow{\text{inward edge normals}} \boxed{\text{normal fan}} \xrightarrow{\text{affine gluing or Cox quotient}} \boxed{\text{toric variety}}.$$

If the question is	Read first	Formula to remember
What is the local chart?	Section 1.2	$U_\sigma = \operatorname{Spec} \mathbb{C}[\sigma^\vee \cap M]$
How do fan charts glue?	Section 1.3	$\tau \preceq \sigma$ gives $U_\tau \subset U_\sigma$
How do I compute the quotient?	Section 1.4	$X_\Sigma = (\mathbb{C}^{\Sigma(1)} \setminus Z_\Sigma)/G$
What geometry can I read off?	Section 1.5	$\dim V(\sigma) = \dim X - \dim \sigma$
How do I start from a polygon?	Section 3.5	$\Delta = \{m : \langle m, v_i\rangle \geq -a_i\}$
How do I get sections?	Section 3.6	$H^0(X, \mathcal{O}(D)) = \bigoplus_{m \in P_D \cap M} \mathbb{C}\chi^m$
What does a blow-up do?	Section 3.2	insert $v_1 + \cdots + v_k$ and subdivide
How do charges enter?	Sections 2.1 and 2.2	$P^{\mathsf{T}} Q = 0$

Common trap: a fan, a set of rays, and a charge matrix are not the same data

The rays and their integer relations determine the Cox group action, but they do not by themselves specify which coordinate sets are forbidden. Different fans can use the same rays, and different GIT chambers can therefore produce different toric varieties. Always record either the cones of the fan, the irrelevant ideal, or the FI/GIT chamber.

Conventions. The lattice of one-parameter subgroups is N and the character lattice is $M = N^\vee$. Polytopes lie in $M_\mathbb{R}$; fans lie in $N_\mathbb{R}$. We use inward facet normals and inequalities $\langle m, v_i\rangle \geq -a_i$. A toric variety is normal unless explicitly stated otherwise. "Complete" is the algebraic geometric term corresponding to compactness over $\mathbb{C}$.

1 Fans: from integral directions to a toric variety

Intuition: why fans describe varieties

On the dense torus $T = (\mathbb{C}^*)^r$, every coordinate is allowed to be nonzero. To compactify or partially compactify T, we decide which one-parameter motions are allowed to acquire limits as $t \to 0$. A lattice vector $u \in N$ represents the motion

$$\lambda^u(t) = (t^{u_1}, \dots, t^{u_r}).$$

A cone σ groups together compatible limiting directions. The dual cone then records exactly the monomials that do not blow up along those directions. This produces one affine chart U_σ. A fan is a compatible collection of cones, so it is precisely the data needed to glue these charts.
The slogan is:

$$\boxed{\text{cone = one affine chart}}, \qquad \boxed{\text{fan = instructions for gluing the charts}}.$$

Definition 1.1. A toric variety X is a complex algebraic variety containing an algebraic torus

$$T = (\mathbb{C}^*)^r$$

as a dense open set, together with an action of T on X whose restriction to $T \subset X$ is just the usual multiplication on T.

The word "toric" therefore does not mean merely that a torus acts. The torus must itself be a dense open orbit. Its complement is the *toric boundary*, built from lower-dimensional torus orbits.

Definition 1.2. A Noetherian integral domain R is called *normal* if it is integrally closed in its field of fractions, i.e. every element of $\mathrm{Frac}(R)$ that is integral over R already lies in R.

Definition 1.3. A variety X is called *normal* if it is irreducible and the local rings $\mathcal{O}_{X,p}$ are normal for all $p \in X$. We will only consider the case that X is a normal variety in this section.

1.1 The two lattices, cones, and fans

Quick lookup: N versus M

	N	$M = N^\vee$
Element	u is a one-parameter subgroup	m is a character/monomial
Map	$\lambda^u : \mathbb{C}^* \to T$	$\chi^m : T \to \mathbb{C}^*$
Coordinates	$t \mapsto (t^{u_1}, \dots, t^{u_r})$	$(z_1, \dots, z_r) \mapsto z_1^{m_1} \cdots z_r^{m_r}$
Geometry stored here	fans, rays, limits	polytopes, exponents, sections

Their pairing is determined by $\chi^m(\lambda^u(t)) = t^{\langle m,u\rangle}$.

Example 1.4. Consider $\mathbb{CP}^r$ with homogeneous coordinates expressed as

$$(x_1, \dots, x_{r+1}).$$

The dense open subset

$$T = \{x : x_i \neq 0,\ i = 1, \dots, r+1\} \subset \mathbb{CP}^r$$

is isomorphic to $(\mathbb{C}^*)^r$ and acts on $\mathbb{CP}^r$ by coordinatewise multiplication, giving $\mathbb{CP}^r$ the structure of a toric variety.

Definition 1.5. A *lattice* is a free abelian group of finite rank, i.e. a group isomorphic to $\mathbb{Z}^r$ for some r.

Let N be a lattice, and set

$$N_{\mathbb{R}} = N \otimes \mathbb{R}.$$

We will denote the rank of N by r. At times, we will fix an isomorphism

$$N \simeq \mathbb{Z}^r,$$

which induces an isomorphism

$$N_{\mathbb{R}} \simeq \mathbb{R}^r.$$

At other times, there will be benefits to thinking of N as an abstract lattice.

Definition 1.6. A *one-parameter subgroup* of a torus T is a morphism

$$\lambda : \mathbb{C}^* \to T$$

that is a group homomorphism.

For example, if $u = (b_1, \dots, b_n) \in \mathbb{Z}^n$, then u defines a one-parameter subgroup

$$\lambda^u : \mathbb{C}^* \to (\mathbb{C}^*)^n$$

given by

$$\lambda^u(t) = (t^{b_1}, \dots, t^{b_n}).$$

All one-parameter subgroups of $(\mathbb{C}^*)^n$ arise in this way, and hence the group of one-parameter subgroups of $(\mathbb{C}^*)^n$ is naturally isomorphic to $\mathbb{Z}^n$.

For an arbitrary torus T, its one-parameter subgroups form a free abelian group N of rank equal to the dimension of T. As with the character group, an element $u \in N$ determines the one-parameter subgroup

$$\lambda^u : \mathbb{C}^* \to T.$$

Definition 1.7. A *character* of a torus T is a morphism

$$\chi : T \to \mathbb{C}^*$$

that is a group homomorphism.

For example, if $m = (a_1, \ldots, a_n) \in \mathbb{Z}^n$, then m defines a character

$$\chi^m : (\mathbb{C}^*)^n \to \mathbb{C}^*$$

given by

$$\chi^m(t_1, \ldots, t_n) = t_1^{a_1} \cdots t_n^{a_n}.$$

All characters of $(\mathbb{C}^*)^n$ arise in this way, and hence the group of characters of $(\mathbb{C}^*)^n$ is naturally isomorphic to $\mathbb{Z}^n$.

For an arbitrary torus T, its characters form a free abelian group M of rank equal to the dimension of T. It is customary to say that an element $m \in M$ gives the character

$$\chi^m : T \to \mathbb{C}^*.$$

We will also need the accompanying vector space

$$M_{\mathbb{R}} = M \otimes \mathbb{R}.$$

Definition 1.8 (Intrinsic pairing)**.** Given a character χ^m and a one-parameter subgroup λ^u, the composition

$$\chi^m \circ \lambda^u : \mathbb{C}^* \to \mathbb{C}^*$$

is a character of $\mathbb{C}^*$, hence is of the form

$$t \longmapsto t^\ell$$

for some $\ell \in \mathbb{Z}$. We define the natural pairing between M and N by

$$\langle m, u \rangle := \ell.$$

It is easy to see M is the dual lattice

$$M = \mathrm{Hom}(T, \mathbb{C}^*) \simeq \mathrm{Hom}(N, \mathbb{Z}).$$

Intuition: the sign of the pairing

Along the path $\lambda^u(t)$, the monomial χ^m becomes $t^{\langle m,u \rangle}$. Hence as $t \to 0$ it

- vanishes if $\langle m, u \rangle > 0$;
- stays invertible if $\langle m, u \rangle = 0$;
- blows up if $\langle m, u \rangle < 0$.

This one observation explains the dual cone, the affine coordinate ring, and the divisor formula $\mathrm{ord}_{D_\rho}(\chi^m) = \langle m, v_\rho \rangle$.

Definition 1.9. A *strongly convex rational polyhedral cone*

$$\sigma \subset N_{\mathbb{R}}$$

is a set of the form

$$\sigma = \{a_1 v_1 + a_2 v_2 + \cdots + a_k v_k \mid a_i \geq 0\},$$

generated by a finite set of vectors $v_1, \ldots, v_k \in N$ such that

$$\sigma \cap (-\sigma) = \{0\}.$$

Definition 1.10 (Faces, rays, and relative interiors)**.** For $m \in \sigma^\vee$, the set

$$\tau = \sigma \cap m^\perp$$

is a *face* of σ, written $\tau \preceq \sigma$. A one-dimensional face is a *ray*. The relative interior $\mathrm{relint}(\sigma)$ is the interior inside the vector space spanned by σ.

Common trap: why strong convexity is imposed

If σ contained a line, then moving in that direction and its negative would both be allowed degenerations. Algebraically, the corresponding chart would carry an extra torus factor. Allowing such cones is possible, but the strongly convex convention gives the cleanest fan–variety dictionary.

Without further comment, strongly convex rational polyhedral cones will simply be referred to as *cones* in this note.

Definition 1.11. A *collection* Σ of strongly convex rational polyhedral cones in $N_\mathbb{R}$ is called a *fan* if

1. each face of a cone in Σ is also a cone in Σ, and
2. the intersection of two cones in Σ is a face of each.

1.2 One cone gives one affine chart

Definition 1.12 (Dual cone and affine toric chart)**.** For a cone $\sigma \subset N_\mathbb{R}$, define

$$\sigma^\vee = \{m \in M_\mathbb{R} \mid \langle m, u\rangle \geq 0 \text{ for every } u \in \sigma\}.$$

The associated affine toric variety is

$$U_\sigma := \operatorname{Spec} \mathbb{C}[\sigma^\vee \cap M].$$

Here $\mathbb{C}[\sigma^\vee \cap M]$ has basis symbols χ^m and multiplication $\chi^m \chi^{m'} = \chi^{m+m'}$.

Recipe: compute U_σ from generators of σ

Suppose $\sigma = \mathrm{Cone}(v_1, \dots, v_k) \subset N_\mathbb{R}$.

1. Write one inequality $\langle m, v_i\rangle \geq 0$ for every generator. Their solution set is $\sigma^\vee \subset M_\mathbb{R}$.
2. Find generators $m_1, \dots, m_s$ of the semigroup $S_\sigma = \sigma^\vee \cap M$ (its Hilbert basis is a minimal choice).
3. Introduce variables $z_j = \chi^{m_j}$. Every integer relation $\sum a_j m_j = \sum b_j m_j$ gives a binomial relation $\prod z_j^{a_j} = \prod z_j^{b_j}$.
4. The result is the affine variety cut out by these binomials.

For a smooth full-dimensional cone, the primitive generators form a lattice basis and the answer is simply $\mathbb{A}^r$.

Example 1.13 (The first quadrant gives $\mathbb{A}^2$)**.** Let $N = \mathbb{Z}^2$ and $\sigma = \mathrm{Cone}(e_1, e_2)$. Then

$$\sigma^\vee = \{(a, b) \in M_\mathbb{R} : a, b \geq 0\}, \qquad \mathbb{C}[\sigma^\vee \cap M] = \mathbb{C}[x, y].$$

Thus $U_\sigma = \mathbb{A}^2$. The open torus is the locus $xy \neq 0$. The two rays correspond to the coordinate axes, and the two-dimensional cone corresponds to their intersection, the origin.

Example 1.14 (A determinant detects a quotient singularity). Let $\sigma = \mathrm{Cone}((1,0),(1,2)) \subset \mathbb{R}^2$. Its primitive generators span a sublattice of index

$$\left|\det\begin{pmatrix}1 & 1\\ 0 & 2\end{pmatrix}\right| = 2.$$

The chart is locally $\mathbb{A}^2/\mathbb{Z}_2$ rather than $\mathbb{A}^2$. In dimension two this determinant test is often the quickest way to detect smoothness.

1.3 A fan glues the affine charts

If $\tau \preceq \sigma$, then U_τ is a torus-invariant open subset of U_σ. For cones $\sigma, \sigma' \in \Sigma$, the fan condition says that $\sigma \cap \sigma'$ is a face of both, so the overlap is

$$U_\sigma \cap U_{\sigma'} = U_{\sigma\cap\sigma'}.$$

Gluing all U_σ along these overlaps produces X_Σ.

Intuition: faces mean localization

Passing from a cone to a face removes one or more boundary conditions. On the ring side, the corresponding monomials are inverted. Thus "take a face" in the fan becomes "take an open subset" in the variety. This reversal is the source of every order-reversing correspondence in toric geometry.

Example 1.15 ($\mathbb{P}^1$ is two copies of $\mathbb{A}^1$ glued along $\mathbb{C}^*$). Take the fan in $N_\mathbb{R} = \mathbb{R}$ with cones

$$\sigma_+ = \mathbb{R}_{\geq 0}, \qquad \sigma_- = \mathbb{R}_{\leq 0}, \qquad \{0\}.$$

Both maximal cones give $\mathbb{A}^1$, while the zero cone gives the torus $\mathbb{C}^*$. Gluing the two affine lines along $\mathbb{C}^*$ by $z \leftrightarrow z^{-1}$ gives $\mathbb{P}^1$. The two rays add the points 0 and ∞ to the torus.

1.4 The Cox quotient: a global coordinate recipe

There are two standard ways to construct a toric variety X_Σ from a fan Σ yielding the same result. The original construction associates an affine toric variety

$$U_\sigma = \mathrm{Spec}\, \mathbb{C}[\sigma^\vee \cap M]$$

to each cone $\sigma \in \Sigma$ and glues them as above.

For global computations it is often more convenient to use Cox homogeneous coordinates. This is the toric analogue of $\mathbb{P}^r = (\mathbb{C}^{r+1} \setminus \{0\})/\mathbb{C}^*$.

Let Σ be a fan in $N_\mathbb{R}$, and let $\Sigma(1)$ denote the set of edges (one-dimensional cones) of Σ. For each $\rho \in \Sigma(1)$, let $v_\rho \in N$ be the unique generator of the semigroup $\rho \cap N$. This vector v_ρ is called the *primitive generator* of ρ. Identifying ρ with v_ρ, the set $\Sigma(1)$ may be regarded as a subset of N.

For ease of exposition, we assume that the vectors v_ρ span $N_\mathbb{R}$ as a vector space for the rest of this chapter.

Let $n = |\Sigma(1)|$. To each edge $\rho \in \Sigma(1)$ we associate a coordinate x_ρ. After choosing an ordering

$$\Sigma(1) = \{v_1, \dots, v_n\},$$

we may write the coordinates as

$$(x_1, \dots, x_n)$$

on $\mathbb{C}^n$.

Recipe: build X_Σ from a fan by Cox coordinates

1. **One ray, one coordinate.** For every primitive ray generator v_i, introduce x_i.
2. **Cones tell which zeros are allowed.** If a set of rays is not contained in a cone, those coordinates are forbidden from vanishing simultaneously. Remove the resulting locus Z_Σ.
3. **Integer relations give rescalings.** Compute

   $$\Lambda = \ker\left(\mathbb{Z}^n \xrightarrow{e_i \mapsto v_i} N\right).$$

   If columns of $Q = (Q_{ia})$ form a basis of the free part of Λ, then $(\lambda_a) \in (\mathbb{C}^*)^{n-r}$ acts by

   $$x_i \longmapsto \left(\prod_a \lambda_a^{Q_{ia}}\right) x_i.$$

   Finite factors must also be retained when the relevant class group has torsion.
4. **Take the quotient.**

   $$X_\Sigma = (\mathbb{C}^n \setminus Z_\Sigma)/G.$$

The rays determine the coordinates and group action; the cones determine the excluded set. Both pieces are essential.

Definition 1.16. Let $S \subset \Sigma(1)$ be a subset which does *not* span a cone of Σ. Define a linear subspace $V(S) \subset \mathbb{C}^n$ by

$$V(S) = \{x_\rho = 0 \text{ for all } \rho \in S\}.$$

Let

$$Z(\Sigma) = \bigcup_S V(S),$$

where the union is taken over all such subsets $S \subset \Sigma(1)$.

Equivalently, it is enough to take the union over minimal forbidden sets. In ring language the same information is encoded by the irrelevant ideal

$$B_\Sigma = \left\langle x^{\widehat{\sigma}} := \prod_{\rho \notin \sigma(1)} x_\rho \;\middle|\; \sigma \in \Sigma \text{ maximal} \right\rangle, \qquad Z_\Sigma = V(B_\Sigma).$$

Definition 1.17. Define a homomorphism

$$\phi\colon \mathrm{Hom}(\Sigma(1),\mathbb{C}^*) \longrightarrow \mathrm{Hom}(M,\mathbb{C}^*)$$

by sending a map of sets $f : \Sigma(1) \to \mathbb{C}^*$ to the group homomorphism

$$m \longmapsto \prod_{v\in\Sigma(1)} f(v)^{\langle m,v\rangle}.$$

In coordinates, suppose that v_j has coordinates

$$(v_{j1},\ldots,v_{jr})$$

with respect to a chosen basis of N (equivalently, the entries are obtained by pairing with the dual basis of M). Then the map ϕ can be written as

$$\phi\colon (\mathbb{C}^*)^n \longrightarrow (\mathbb{C}^*)^r, \qquad (t_1,\ldots,t_n) \longmapsto \left(\prod_{j=1}^n t_j^{v_{j1}},\ldots,\prod_{j=1}^n t_j^{v_{jr}}\right).$$

Definition 1.18. The group G is defined to be the kernel of ϕ:

$$G = \mathrm{Ker}\left(\mathrm{Hom}(\Sigma(1),\mathbb{C}^*) \xrightarrow{\phi} \mathrm{Hom}(M,\mathbb{C}^*)\right).$$

Since $G \subset \mathrm{Hom}(\Sigma(1),\mathbb{C}^*)$, for each $g \in G$ and each $\rho \in \Sigma(1)$ we have $g(v_\rho) \in \mathbb{C}^*$. This defines an action of G on $\mathbb{C}^n$ by

$$g\cdot(x_1,\ldots,x_n) = (g(v_1)x_1,\ldots,g(v_n)x_n).$$

It is easy to see that this action preserves $\mathbb{C}^n - Z(\Sigma)$.

Definition 1.19. The toric variety associated to the fan Σ is defined by

$$X_\Sigma = (\mathbb{C}^n - Z(\Sigma))/G.$$

For a simplicial fan this is a geometric quotient in the usual orbit-space sense. For a general fan one should interpret it as the Cox good quotient; orbit closures can meet, so the naive set of orbits is not the correct algebraic variety.

The toric variety X_Σ contains the dense open torus

$$T = (\mathbb{C}^*)^n/G,$$

which acts on X_Σ by coordinatewise multiplication. It is easy to see that this torus has rank r, so that X_Σ is an r-dimensional toric variety.

Example 1.20 (Run the entire recipe for $\mathbb{P}^2$)**.** Take the rays

$$v_1 = (1,0), \qquad v_2 = (0,1), \qquad v_3 = (-1,-1),$$

and take every cone generated by a proper subset of adjacent rays.

1. The rays give coordinates (x_1,x_2,x_3).
2. The only minimal forbidden set is $\{v_1,v_2,v_3\}$, hence $Z_\Sigma = \{0\}$.
3. The only relation is $v_1+v_2+v_3 = 0$, hence $\lambda\cdot(x_1,x_2,x_3) = (\lambda x_1,\lambda x_2,\lambda x_3)$.
4. Therefore
$$X_\Sigma = (\mathbb{C}^3 \setminus \{0\})/\mathbb{C}^* = \mathbb{P}^2.$$

This example is the template for reading a fan as generalized homogeneous coordinates.

1.5 The fan–geometry dictionary

Quick lookup: what to read directly from a fan

Let $r = \dim X_\Sigma$.

Combinatorics	Geometry
zero cone $\{0\}$	dense torus T and its closure X_Σ
ray ρ	invariant prime divisor D_ρ
k**-cone** σ	orbit $O(\sigma)$ of dimension $r-k$ and closure $V(\sigma)$
maximal r**-cone**	torus fixed point
$\tau \preceq \sigma$	$V(\sigma) \subseteq V(\tau)$
$\lvert\Sigma\rvert = N_\mathbb{R}$	X_Σ is complete
every cone unimodular	X_Σ is smooth
every cone simplicial	X_Σ is $\mathbb{Q}$-factorial; locally a finite quotient
subdivision of Σ	proper birational toric morphism when supports agree

Definition 1.21. Let Σ be a fan and let $\sigma \in \Sigma$ be a cone generated by edges $\rho_1, \ldots, \rho_k$. The associated T-invariant subvariety is defined by

$$Z_\sigma = V(\sigma) = \{\, x \in X_\Sigma \mid x_{\rho_1} = \cdots = x_{\rho_k} = 0 \,\},$$

where x_ρ are the homogeneous coordinates of x.

Proposition 1.22 (Classification of T-invariant subvarieties)**.** *The assignment*

$$\sigma \longmapsto Z_\sigma$$

defines an order-reversing bijection between cones in the fan Σ and non-empty T-invariant subvarieties of X_Σ.

Remark 1.23. In particular, the edges of Σ are in one-to-one correspondence with T-invariant divisors in X_Σ. More generally, if σ is a k-dimensional cone, then Z_σ is an $(r-k)$-dimensional subvariety of X_Σ.

Remark 1.24. If a set of edges $\{\rho_1, \ldots, \rho_\ell\}$ does not span a cone in Σ, then the equations

$$x_{\rho_1} = \cdots = x_{\rho_\ell} = 0$$

define a subset of $\mathbb{C}^n$ contained in $Z(\Sigma)$, hence they define the empty set in X_Σ.

Remark 1.25. Each Z_σ is itself a toric variety. Its fan is constructed by replacing N with the quotient lattice

$$N' = N/\langle \sigma \cap N \rangle,$$

and projecting each cone in Σ which contains σ as a face to N'. The resulting collection of cones forms a fan in N'.

At this point, we relate the homogeneous coordinate construction with the standard affine covering

$$X_\Sigma = \bigcup_{\sigma \in \Sigma} X_\sigma.$$

Definition 1.26. For $\sigma \in \Sigma$, define the Cox open set

$$\widehat{U}_\sigma = \{x \in \mathbb{C}^n \setminus Z_\Sigma : x^{\widehat{\sigma}} \neq 0\} = \{x_\rho \neq 0 \text{ for every } \rho \notin \sigma(1)\}.$$

Its quotient is the affine chart

$$\widehat{U}_\sigma / G \simeq U_\sigma = \operatorname{Spec} \mathbb{C}[\sigma^\vee \cap M].$$

Remark 1.27. In simple examples one can use the G-action to choose representatives with $x_\rho = 1$ for $\rho \notin \sigma(1)$. This is a convenient gauge choice, not the intrinsic definition of the chart, and it may have residual finite ambiguity. The nonvanishing condition above is the safe statement to use.

1.6 Compactness Criterion and Smoothness Criterion

Proposition 1.28 (Compactness criterion for toric varieties)**.** *Let X_Σ be a toric variety. Then the following are equivalent:*

1. *X_Σ is compact in the classical topology.*
2. *The fan Σ is complete, i.e.,*

$$|\Sigma| = \bigcup_{\sigma \in \Sigma} \sigma = N_\mathbb{R}.$$

Proposition 1.29 (Smoothness criterion for toric varieties)**.** *A toric variety X_Σ is smooth if and only if every cone $\sigma \in \Sigma$ is generated by a $\mathbb{Z}$-basis of the lattice*

$$\operatorname{Span}_\mathbb{R}(\sigma) \cap N.$$

Proof. If the primitive generators of σ extend to a basis of N, then after a lattice change of coordinates

$$U_\sigma \simeq \mathbb{A}^{\dim \sigma} \times (\mathbb{C}^*)^{r - \dim \sigma},$$

so every chart is smooth. Conversely, the distinguished closed orbit in U_σ is smooth only if the semigroup $\sigma^\vee \cap M$ is freely generated in the transverse directions, which is equivalent to the primitive ray generators of σ extending to a lattice basis. Since the U_σ cover X_Σ, the criterion follows. □

Quick lookup: the two-dimensional tests

Write the primitive rays of a complete surface fan counterclockwise as $v_1, \ldots, v_n$ (cyclic indices).

- Smoothness is equivalent to $\det(v_i, v_{i+1}) = 1$ for every i.
- If the surface is smooth, there is a unique integer a_i with $v_{i-1} + v_{i+1} = a_i v_i$, and the corresponding invariant curve has self-intersection $D_i^2 = -a_i$.
- Blowing up the fixed point for $\operatorname{Cone}(v_i, v_{i+1})$ inserts the ray generated by $v_i + v_{i+1}$.

These three calculations handle a large fraction of day-to-day toric surface problems.

1.7 Worked example: the local projective plane

Intuition: how a line bundle appears in Cox coordinates

The first three coordinates behave like homogeneous coordinates on $\mathbb{P}^2$. The fourth is a fiber coordinate. Rescaling the base coordinates by t forces the fiber coordinate to rescale by t^{-3}, so the fiber transforms with degree -3. This is exactly the transition behavior of $\mathcal{O}_{\mathbb{P}^2}(-3)$.

Example 1.30. We construct the total space $\mathcal{O}_{\mathbb{CP}^2}(-3)$ from a fan Σ in $\mathbb{R}^3$. Set

$$\Sigma(1) = \{(1,0,1),\ (0,1,1),\ (-1,-1,1),\ (0,0,1)\}.$$

The convex hull of $\Sigma(1)$ is a triangle in the plane $z = 1$ with vertices $\{v_1, v_2, v_3\}$; the vector v_4 lies in the interior of this triangle, subdividing it into three smaller triangles. The three-dimensional cones in Σ are the cones over these triangles, and the remaining cones are their faces. These cones do not span $\mathbb{R}^3$, which is consistent with the fact that $\mathcal{O}_{\mathbb{CP}^2}(-3)$ is non-compact (non-compact toric varieties of this type are useful in geometric engineering).

We compute that

$$Z(\Sigma) = \{x_1 = x_2 = x_3 = 0\}.$$

Moreover, the group G is the kernel of the homomorphism

$$\phi\colon (\mathbb{C}^*)^4 \longrightarrow (\mathbb{C}^*)^3, \qquad (t_1, t_2, t_3, t_4) \longmapsto (t_1 t_3^{-1},\ t_2 t_3^{-1},\ t_1 t_2 t_3 t_4),$$

so that

$$G = \{(t, t, t, t^{-3}) \mid t \in \mathbb{C}^*\} \simeq \mathbb{C}^*.$$

There are four T-invariant divisors $D_1, \ldots, D_4$. Since $\{v_1, v_2, v_3\}$ does not span a cone in Σ, the divisors D_1, D_2, D_3 have empty intersection. All other triples of divisors intersect in a point, since the corresponding edges span a three-dimensional cone. The toric description of the D_i shows that D_4 is compact, while the other divisors are non-compact.

Projection to the first three factors defines a morphism

$$\mathrm{X}_\Sigma \longrightarrow \mathbb{CP}^2$$

whose fibers are isomorphic to $\mathbb{C}$. Hence X_Σ is a line bundle over $\mathbb{CP}^2$, as claimed. The divisor D_4 is identified with the zero section of the bundle, and the divisors D_1, D_2, D_3 are the restrictions of this bundle over the coordinate lines $x_1 = 0$, $x_2 = 0$, $x_3 = 0$.

1.8 Constructing Fans from Toric Varieties

Intuition: recovering the fan by asking which limits exist

For each lattice direction $u \in N$, send t toward zero along $\lambda^u(t)$. Directions that land in the same torus orbit belong to the relative interior of the same cone. Directions with no limit lie outside the support of the fan. Thus the fan is a “limit map” for all one-parameter subgroups at once.

Let X be a normal toric variety containing the torus

$$T \simeq (\mathbb{C}^*)^r.$$

We consider the lattice

$$N = \mathrm{Hom}(\mathbb{C}^*, T) \simeq \mathbb{Z}^r,$$

whose elements are one-parameter subgroups of T. Identifying T with $(\mathbb{C}^*)^r$, this identification may be written as

$$\mathbb{Z}^r \simeq N, \qquad (a_1, \dots, a_r) \longmapsto \big(t \mapsto (t^{a_1}, \dots, t^{a_r})\big).$$

For $u \in N$, consider

$$\lambda^u : \mathbb{C}^* \longrightarrow T \subset X, \qquad t \longmapsto \lambda^u(t) \cdot 1.$$

Proposition 1.31 (One-parameter subgroup test). *For $X = X_\Sigma$ the limit $\lim_{t\to 0} \lambda^u(t)$ exists if and only if $u \in |\Sigma|$. More precisely, if $u \in \mathrm{relint}(\sigma)$, then the limit lies in the orbit $O(\sigma)$ and*

$$\overline{T \cdot \lim_{t\to 0} \lambda^u(t)} = V(\sigma).$$

Thus the relative interiors of the cones are exactly the equivalence classes of directions having limits in the same torus orbit.

Conversely, starting with a normal toric variety X, partition the set of one-parameter subgroups for which the limit exists according to the orbit containing the limit. Taking the real closures of these regions recovers a fan Σ, and the classification theorem gives $X \simeq X_\Sigma$.

Remark 1.32 (Compactness criterion revisited). This construction gives a conceptual explanation of the compactness criterion for toric varieties. If

$$|\Sigma| \subsetneq N_\mathbb{R},$$

then there exists a one-parameter subgroup $\psi \in N$ not contained in any cone of Σ. For such a ψ, the limit

$$\lim_{t\to 0} \psi(t)$$

does not exist in X_Σ, and hence X_Σ cannot be compact.

Example 1.33 (Recovering the fan of $\mathbb{CP}^2$ from one-parameter subgroups). We revisit $\mathbb{CP}^2$ as a toric variety and recover its fan using the construction of one-parameter subgroups.

Using the rescaling of homogeneous coordinates, we may normalize the first coordinate to 1, identifying the dense torus as

$$T = \{(1, t_1, t_2) \mid t_i \in \mathbb{C}^*\} \simeq (\mathbb{C}^*)^2.$$

With this identification, the torus action on $\mathbb{CP}^2$ is given by

$$(t_1, t_2) \cdot (x_1, x_2, x_3) = (x_1, t_1 x_2, t_2 x_3).$$

The one-parameter subgroups of T are indexed by $(a, b) \in \mathbb{Z}^2$ and are given by

$$\psi_{a,b}(t) = (t^a, t^b) \in N = \mathrm{Hom}(\mathbb{C}^*, T).$$

Using the embedding $T \subset \mathbb{CP}^2$, we may study the limit

$$\lim_{t\to 0} \psi_{a,b}(t) \in \mathbb{CP}^2.$$

Condition on (a,b)	$\lim_{t\to 0}\psi(t)$	orbit closure
$a>0,\ b>0$	$(1,0,0)$	$\{(1,0,0)\}$
$a<0,\ b>a$	$(0,1,0)$	$\{(0,1,0)\}$
$b<0,\ b<a$	$(0,0,1)$	$\{(0,0,1)\}$
$a=b<0$	$(0,1,1)$	$\{x_1=0\}$
$a>0,\ b=0$	$(1,0,1)$	$\{x_2=0\}$
$a=0,\ b>0$	$(1,1,0)$	$\{x_3=0\}$
$a=b=0$	$(1,1,1)$	$\mathbb{CP}^2$

Table 1: Limits of one-parameter subgroups in $\mathbb{CP}^2$

There are seven possibilities for the limit points and their orbit closures, summarized in Table 1. The closures of the regions defined by the conditions on (a,b) determine the cones of the fan of $\mathbb{CP}^2$. In this way, we recover the fan consisting of seven cones, in agreement with the standard toric description of $\mathbb{CP}^2$. This also recovers the correspondence between cones and non-empty T-invariant subvarieties.

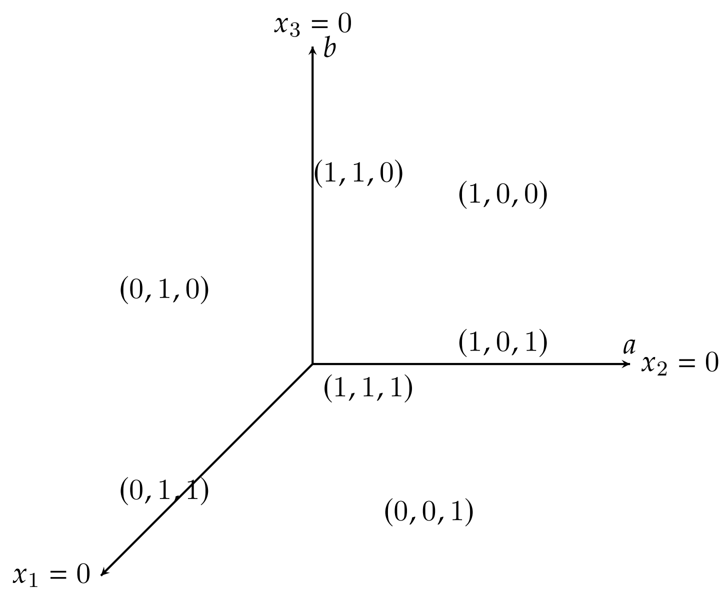

Figure 1: One-parameter subgroups and limit points for $\mathbb{CP}^2$

2 GLSM and Intersection Numbers

The *gauged linear sigma model* (GLSM) is a two-dimensional gauge theory. For present purposes, we restrict our attention to GLSMs without a superpotential.

2.1 Supersymmetric Ground States

Intuition: the same quotient in symplectic and holomorphic language

The D-term equations choose a level of a moment map and $U(1)^s$ removes phase directions. After complexifying the gauge group, the same space is obtained by removing unstable points and quotienting by $(\mathbb{C}^*)^s$. The charge matrix is therefore the Cox weight matrix; the FI parameter chooses which sets of coordinates are allowed to vanish, hence which fan/GIT chamber is selected.

Definition 2.1 (GLSM without superpotential)**.** We consider a two-dimensional $\mathcal{N} = (2, 2)$ gauged linear sigma model (GLSM) with gauge group $U(1)^s$, vector superfields $V_1, \dots, V_s$, and n chiral superfields $\Phi_1, \dots, \Phi_n$.

The charge of Φ_i under the a-th $U(1)$ factor is denoted by $Q_{i,a} \in \mathbb{Z}$, and the scalar component of Φ_i is denoted by ϕ_i. The Lagrangian takes the form

$$L = L_{\text{kin}} + L_{\text{gauge}} + L_{D,\theta},$$

where the three terms correspond respectively to the kinetic energy of the chiral fields, the kinetic energy of the gauge fields, and the Fayet–Iliopoulos (FI) and theta terms.

The scalar potential is given by

$$U(\phi) = \sum_{a=1}^{s} \frac{e_a^2}{2} \left(\sum_{i=1}^{n} Q_{i,a} |\phi_i|^2 - r_a \right)^2,$$

where e_a are the gauge couplings and $r_a \in \mathbb{R}$ are the FI parameters.

The supersymmetric ground states are defined as the solutions of the D-term equations

$$\sum_{i=1}^{n} Q_{i,a} |\phi_i|^2 = r_a, \qquad a = 1, \dots, s,$$

modulo the action of the gauge group $U(1)^s$.

Theorem 2.2 (Vacuum moduli space as a toric variety)**.** *Suppose the charge map has rank s and choose an FI parameter in a geometric GIT chamber. If the resulting stable locus is nonempty and the quotient has the expected dimension, the vacuum moduli space is a normal toric variety of dimension $n - s$. When the coordinate weights correspond to distinct primitive nonzero ray generators, its fan has n rays. Repeated, nonprimitive, or zero weights require the stacky or non-effective version of this statement.*

Proof. Define a subgroup $G = (\mathbb{C}^*)^s \subset (\mathbb{C}^*)^n$ by the embedding

$$(t_1, \dots, t_s) \longmapsto \left(\prod_{a=1}^{s} t_a^{Q_{1,a}}, \dots, \prod_{a=1}^{s} t_a^{Q_{n,a}} \right).$$

The residual torus acting effectively on the vacuum moduli space is

$$T = (\mathbb{C}^*)^n / G.$$

Let

$$N = \mathrm{Hom}(\mathbb{C}^*, T)$$

be the lattice of one-parameter subgroups of the torus T. There exists a collection of elements

$$S = \{v_1, \ldots, v_n\} \subset N$$

such that replacing the set $\Sigma(1)$ of rays in the Cox construction by S reproduces the embedding. For simplicity, we assume the v_i are distinct and set

$$\Sigma(1) := S.$$

Consider subsets

$$P = \{v_{i_1}, \ldots, v_{i_k}\} \subset \Sigma(1)$$

such that there are no solutions to the D-term equations

$$\sum_i Q_{i,a}|\phi_i|^2 = r_a$$

with $\phi_{i_1} = \cdots = \phi_{i_k} = 0$.

Within a fixed geometric chamber, the unstable coordinate subspaces determine the irrelevant ideal and hence the cones of a fan Σ. The resulting GIT quotient is the Cox construction of X_Σ. □

Remark 2.3. We explain how the gauge charges in a GLSM determine the toric variety of supersymmetric ground states, and conversely how the toric data recover the GLSM charges.

One associates to the GLSM a set of vectors

$$\Sigma(1) = \{v_1, \ldots, v_n\}$$

which generate the one-dimensional cones of a fan Σ. By construction, the charges encode the linear relations among the v_i:

$$\sum_{i=1}^{n} Q_{i,a} v_i = 0, \qquad a = 1, \ldots, s.$$

Conversely, starting from the set of toric rays $\Sigma(1)$, one considers the lattice

$$\Lambda := \left\{ (q_1, \ldots, q_n) \in \mathbb{Z}^n \;\middle|\; \sum_{i=1}^{n} q_i v_i = 0 \right\},$$

which is the lattice of all $\mathbb{Z}$-linear relations among the v_i. This lattice has rank s.

Choosing a basis of Λ, one obtains relations

$$\sum_{i=1}^{n} Q'_{i,a} v_i = 0, \qquad a = 1, \ldots, s,$$

and hence a charge matrix $Q'_{i,a}$.

From linear algebra, the coefficients $Q'_{i,a}$ are precisely the charges of the original chiral superfields Φ_i, with the understanding that different choices of basis of Λ correspond to different ways of writing the same gauge group as a product of s copies of $U(1)$.

Remark 2.4 (Kempf–Ness correspondence for GLSM)**.** In the GLSM construction, the space of supersymmetric ground states is defined as the Kähler quotient

$$\mathcal{M}_{\text{vac}} = \mu^{-1}(r)/U(1)^s,$$

where $\mu : \mathbb{C}^n \to \mathbb{R}^s$ is the moment map

$$\mu_a(\phi) = \sum_{i=1}^{n} Q_{i,a}|\phi_i|^2,$$

and $r = (r_1, \ldots, r_s)$ are the FI parameters. Note that the complexified gauge group $G = (\mathbb{C}^*)^s$ does *not* preserve the level set $\mu^{-1}(r)$.

Nevertheless, one can equivalently describe the vacuum moduli space as a GIT quotient

$$\mathcal{M}_{\text{vac}} \;\cong\; (\mathbb{C}^n \setminus Z_r)/G,$$

where Z_r is the unstable locus determined by the FI parameters. This equivalence is a special case of the Kempf–Ness correspondence, which asserts that every polystable G-orbit intersects $\mu^{-1}(r)$ in a unique $U(1)^s$-orbit. More generally, each GIT equivalence class contains a unique closed orbit, and that orbit has such an intersection. In the common chamber where stable equals semistable, this reduces to the simpler orbit-by-orbit statement often used in GLSM calculations.

Thus, although the G-action does not preserve $\mu^{-1}(r)$, the Kähler quotient by $U(1)^s$ is canonically identified with the GIT quotient by G. This identification allows one to construct the vacuum moduli space as a toric variety using the holomorphic quotient $(\mathbb{C}^n \setminus Z_r)/G$.

Remark 2.5. The space of supersymmetric ground states depends crucially on the choice of FI parameters r_a. For certain values of r_a, the vacuum moduli space may fail to be a toric variety, or may even be empty.

For example, consider a $U(1)$ gauge theory with two chiral fields of charges $(1, 1)$. The D-term equation

$$|\phi_1|^2 + |\phi_2|^2 = r$$

has no solutions if $r < 0$, so in this regime the space of supersymmetric ground states is empty. This shows that the existence of a geometric phase already imposes nontrivial constraints on the FI parameters.

More generally, the dependence of the vacuum geometry on the FI parameters is encoded in the *GKZ decomposition* of the parameter space, which describes a chamber structure corresponding to different geometric and non-geometric phases of the GLSM. Crossing a wall in the GKZ fan corresponds to a birational transformation of the target space, such as a blow-up or flop.

Finally, we note that for a general toric variety the group G appearing in the GLSM construction need not be a pure torus $(\mathbb{C}^*)^s$; it may also contain finite abelian factors. In such cases, the resulting vacuum moduli space is a toric *orbifold*, rather than a smooth toric variety. Producing these geometries requires allowing orbifold phases in the GLSM, which will be discussed later.

Example 2.6 (A $U(1)$ GLSM with charges $(1, 1, 1, -3)$)**.** Consider a two-dimensional $U(1)$ gauge theory with four chiral fields $\Phi_1, \Phi_2, \Phi_3, \Phi_4$ of charges

$$Q = (1, 1, 1, -3).$$

Let $r \in \mathbb{R}$ be the FI parameter. The D-term (moment map) equation is

$$|\phi_1|^2 + |\phi_2|^2 + |\phi_3|^2 - 3|\phi_4|^2 = r,$$

where ϕ_i denotes the scalar component of Φ_i.

(i) The phase $r > 0$. If $r > 0$, then D-term equation implies that we cannot have $\phi_1 = \phi_2 = \phi_3 = 0$. Equivalently, in the holomorphic/GIT picture, the unstable locus contains $\{\phi_1 = \phi_2 = \phi_3 = 0\}$, hence the quotient removes this locus. In this chamber one obtains the geometric phase whose vacuum moduli space is the total space of the line bundle

$$\mathrm{Tot}\big(\mathcal{O}_{\mathbb{P}^2}(-3)\big).$$

(Concretely, this is the toric variety described by the fan in the earlier toric example, where the excluded set is $Z = \{x_1 = x_2 = x_3 = 0\}$.)

(ii) The phase $r < 0$ and the orbifold $\mathbb{C}^3/\mathbb{Z}_3$. If $r < 0$, then D-term equation forces $\phi_4 \neq 0$ (otherwise the left-hand side is ≥ 0). This has two closely related consequences: *(a) Gauge-fixing and the residual $\mathbb{Z}_3$.* The $U(1)$ action is

$$(\phi_1, \phi_2, \phi_3, \phi_4) \longmapsto (e^{i\theta}\phi_1,\ e^{i\theta}\phi_2,\ e^{i\theta}\phi_3,\ e^{-3i\theta}\phi_4).$$

Since $\phi_4 \neq 0$, we may use the continuous $U(1)$-freedom to fix the phase of ϕ_4 (and in the holomorphic picture, to set $\phi_4 = 1$). However, the gauge transformation with $\theta = 2\pi/3$ preserves this gauge choice because $e^{-3i\theta} = e^{-2\pi i} = 1$. Therefore, after gauge-fixing there remains a *finite residual gauge group*

$$\mathbb{Z}_3 = \{e^{2\pi i k/3} \mid k = 0, 1, 2\} \subset U(1).$$

This residual $\mathbb{Z}_3$ acts on (ϕ_1, ϕ_2, ϕ_3) by simultaneous multiplication by a primitive cube root of unity:

$$(\phi_1, \phi_2, \phi_3) \longmapsto (\omega\phi_1, \omega\phi_2, \omega\phi_3), \qquad \omega = e^{2\pi i/3}.$$

Hence the moduli space in this phase is

$$\mathbb{C}^3/\mathbb{Z}_3.$$

(b) The same statement in the holomorphic/GIT quotient language. Consider the holomorphic $(\mathbb{C}^*)$-action with weights $(1, 1, 1, -3)$:

$$\lambda \cdot (\phi_1, \phi_2, \phi_3, \phi_4) = (\lambda\phi_1, \lambda\phi_2, \lambda\phi_3, \lambda^{-3}\phi_4).$$

For $r < 0$, stability forces $\phi_4 \neq 0$, so on the stable locus we set $\phi_4 = 1$ by choosing $\lambda = \phi_4^{1/3}$. The ambiguity of the cube root is exactly a $\mathbb{Z}_3$-freedom, producing the same quotient $\mathbb{C}^3/\mathbb{Z}_3$.

(iii) Relation between the two phases. Crossing $r = 0$ changes the GIT stability condition (a wall-crossing in the GKZ/secondary fan). Geometrically, the $r > 0$ phase gives the smooth resolution $\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^2}(-3))$, while the $r < 0$ phase is the orbifold $\mathbb{C}^3/\mathbb{Z}_3$; the former can be viewed as a (crepant) resolution of the latter.

Definition 2.7 (The $P \mid Q$ matrix)**.** Let $\Sigma(1) = \{v_1, \ldots, v_n\}$ be the set of generators of the one-dimensional cones of a toric fan, and let

$$\Lambda := \left\{ (q_1, \ldots, q_n) \in \mathbb{Z}^n \;\middle|\; \sum_{i=1}^{n} q_i v_i = 0 \right\}$$

be the lattice of $\mathbb{Z}$-linear relations among the v_i, of rank s.

Choosing a basis of the lattice N and a basis of Λ, the toric data can be encoded in a matrix

$$P \mid Q,$$

where the matrix P has row vectors given by the generators v_i of the toric rays; the matrix Q has column vectors forming a basis of the relation lattice Λ.

Equivalently, the matrix Q satisfies the relations

$$\sum_{i=1}^{n} Q_{i,a} v_i = 0, \qquad a = 1, \dots, s.$$

With this row convention,

$$P \in \mathrm{Mat}_{n\times r}(\mathbb{Z}), \qquad Q \in \mathrm{Mat}_{n\times s}(\mathbb{Z}), \qquad P^{\mathsf{T}} Q = 0.$$

Recipe: move between rays and charges

- From rays to charges: compute the integer kernel of P^{T}; put a saturated basis into the columns of Q.
- From charges to rays: compute the Gale dual integer cokernel, then choose primitive ray representatives. Finally specify a GIT chamber; Q alone does not determine the fan.
- Check the answer by verifying $P^{\mathsf{T}} Q = 0$ and by comparing the forbidden coordinate sets with the intended fan.

Example 2.8 (The Hirzebruch surface F_n as a GLSM vacuum)**.** Consider the Hirzebruch surface F_n. The lattice of $\mathbb{Z}$-linear relations among the toric rays is generated by the matrix

$$P \mid Q = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ -1 & -n & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & -n \end{array}\right).$$

It follows that this toric variety can be realized as the space of supersymmetric ground states of a GLSM with gauge group $U(1)^2$ and four chiral superfields. The corresponding charge vectors are

$$(0,1), \quad (0,1), \quad (1,0), \quad (1,-n),$$

which can be checked directly from the relation lattice.

2.2 Intersection Numbers and Charges

Intuition: why a charge is an intersection number

A row of Q tells how the Cox coordinate x_i transforms, hence which divisor class $D_i = \{x_i = 0\}$ represents. A column chooses a relation/curve direction. Pairing the divisor class with that curve produces the same integer. The identity $Q_{ia} = D_i \cdot C_a$ is therefore a change of language, not new data.

Definition 2.9 (T-invariant divisors)**.** Let X_Σ be a toric variety with fan Σ. For each $\rho \in \Sigma(1)$, denote by D_ρ the T-invariant divisor corresponding to the ray ρ. Then the group of T-invariant Weil divisors is

$$\mathbb{Z}^{\Sigma(1)} \cong \bigoplus_{\rho\in\Sigma(1)} \mathbb{Z} \cdot D_\rho.$$

Definition 2.10 (Principal divisors from characters)**.** Let M be the character lattice of the torus T. Each character $m \in M$ defines a rational function on X_Σ, whose zeros and poles determine a principal divisor

$$(m) = \sum_{\rho\in\Sigma(1)} \langle m, v_\rho \rangle D_\rho.$$

This defines a homomorphism

$$M \longrightarrow \mathbb{Z}^{\Sigma(1)}.$$

This definition above is just equivalent to the usual divisor we use in algebraic geometry. We will quickly prove this.

Lemma 2.11 (Generic point lies in a toric affine chart)**.** *Let X_Σ be a toric variety, $\rho \in \Sigma(1)$, and $\sigma \in \Sigma$ such that $\rho \subset \sigma$. Let $D_\rho \subset X_\Sigma$ be the torus-invariant prime divisor corresponding to ρ, and let η_ρ be its generic point. Then $\eta_\rho \in U_\sigma$ and*

$$\mathcal{O}_{X_\Sigma,\eta_\rho} \cong \mathcal{O}_{U_\sigma,\eta_\rho}.$$

Proof. Since $\rho \subset \sigma$, the orbit–face correspondence implies that $D_\rho \cap U_\sigma$ is the orbit closure $V(\rho) \subset U_\sigma$, hence it is a nonempty irreducible closed subset of U_σ. The generic point η_ρ of D_ρ belongs to every nonempty open subset of D_ρ; in particular, it lies in the nonempty open subset $D_\rho \cap U_\sigma \subset D_\rho$. Therefore $\eta_\rho \in U_\sigma$. The identification of local rings follows from the general fact that for an open immersion $j : U_\sigma \hookrightarrow X_\Sigma$ and a point $x \in U_\sigma$, one has $\mathcal{O}_{X_\Sigma,x} \cong \mathcal{O}_{U_\sigma,x}$. □

Lemma 2.12 (The divisor D_ρ on an affine chart)**.** *Let $\sigma \in \Sigma$ be a cone and $\rho \preceq \sigma$ a ray. Write $U_\sigma =$ Spec $\mathbb{C}[\sigma^\vee \cap M]$. Then the closed subscheme $D_\rho \cap U_\sigma \subset U_\sigma$ is defined by the prime ideal*

$$\mathfrak{p}_\rho(\sigma) := \left(\chi^u \;\middle|\; u \in \sigma^\vee \cap M,\ \langle u, v_\rho \rangle > 0\right) \subset \mathbb{C}[\sigma^\vee \cap M].$$

In particular, $D_\rho \cap U_\sigma$ is nonempty and irreducible.

Proof. This is the standard orbit–face description of toric subvarieties: for a face $\tau \preceq \sigma$, the orbit closure $V(\tau) \subset U_\sigma$ is defined by the ideal generated by monomials χ^u such that $\langle u, v \rangle > 0$ for some (equivalently any) $v \in \tau$ in the relative interior. For $\tau = \rho$ this yields the stated ideal. □

Proposition 2.13 (Local coordinate for D_ρ in the smooth case)**.** *Assume X_Σ is smooth. Let $\sigma \in \Sigma$ be a maximal cone of dimension r with rays $\rho_1, \dots, \rho_r$ and primitive generators $v_{\rho_1}, \dots, v_{\rho_r} \in N$. Then $\{v_{\rho_i}\}$ is a $\mathbb{Z}$-basis of N, and letting $\{u_i\} \subset M$ be the dual basis $\langle u_i, v_{\rho_j} \rangle = \delta_{ij}$, we have an identification*

$$\mathbb{C}[\sigma^\vee \cap M] \cong \mathbb{C}[\chi^{u_1}, \dots, \chi^{u_r}] \cong \mathbb{C}[x_1, \dots, x_r], \qquad x_i := \chi^{u_i}.$$

Under this identification, for each j,

$$D_{\rho_j} \cap U_\sigma = V(x_j), \qquad \text{and} \qquad \mathcal{O}_{U_\sigma,\eta_{\rho_j}} \cong \mathbb{C}[x_1,\ldots,x_r]_{(x_j)},$$

so x_j is a uniformizer of the DVR $\mathcal{O}_{X_\Sigma,\eta_{\rho_j}}$.

Proof. Smoothness implies σ is unimodular, hence the primitive ray generators of a maximal cone form a $\mathbb{Z}$-basis of N and admit a dual basis in M. Then $\sigma^\vee \cap M$ is the free commutative monoid generated by $u_1,\ldots,u_r$, so the semigroup algebra is a polynomial algebra. The description of $D_{\rho_j} \cap U_\sigma$ as $V(x_j)$ follows by comparing the ideal $\mathfrak{p}_{\rho_j}(\sigma)$ from the previous lemma with the condition $\langle u_i, v_{\rho_j}\rangle = \delta_{ij}$: a monomial $\chi^u = \prod_i x_i^{a_i}$ lies in $\mathfrak{p}_{\rho_j}(\sigma)$ iff $\langle u, v_{\rho_j}\rangle = a_j > 0$, i.e. iff it is divisible by x_j, so $\mathfrak{p}_{\rho_j}(\sigma) = (x_j)$. Localizing at the generic point gives the stated DVR. □

Theorem 2.14 (Valuation of a character along D_ρ)**.** *Let X_Σ be a smooth toric variety. For any $m \in M$ and any ray $\rho \in \Sigma(1)$,*

$$\mathrm{ord}_{D_\rho}(\chi^m) = \langle m, v_\rho\rangle.$$

Consequently, the principal divisor of χ^m is

$$\div(\chi^m) = \sum_{\rho\in\Sigma(1)} \langle m, v_\rho\rangle D_\rho.$$

Proof. Fix a maximal cone σ containing ρ (such a σ exists) and write $\rho = \rho_j$ in the notation of the previous proposition. By Definition 2.11, $\eta_\rho \in U_\sigma$ and $\mathcal{O}_{X_\Sigma,\eta_\rho} \cong \mathcal{O}_{U_\sigma,\eta_\rho}$. By the proposition, we may identify

$$\mathcal{O}_{U_\sigma,\eta_\rho} \cong \mathbb{C}[x_1,\ldots,x_r]_{(x_j)}$$

with uniformizer $x_j = \chi^{u_j}$. Writing $m = \sum_{i=1}^r a_i u_i$ in the dual basis, we have

$$\chi^m = \prod_{i=1}^r (\chi^{u_i})^{a_i} = \prod_{i=1}^r x_i^{a_i}.$$

In the DVR $\mathbb{C}[x_1,\ldots,x_r]_{(x_j)}$, all x_i with $i \neq j$ are units, hence the valuation $\mathrm{ord}_{D_\rho}(\chi^m)$ equals the exponent of the uniformizer x_j, namely a_j. By duality, $a_j = \langle m, v_{\rho_j}\rangle = \langle m, v_\rho\rangle$. This proves the first formula. The expression for $\div(\chi^m)$ follows from the definition

$$\div(f) = \sum_{\rho\in\Sigma(1)} \mathrm{ord}_{D_\rho}(f)\, D_\rho$$

applied to $f = \chi^m$. □

Remark 2.15. For $v \in N$ let $\lambda_v : \mathbb{C}^* \to T$ be the associated one-parameter subgroup, and for $m \in M$ let $\chi^m : T \to \mathbb{C}^*$ be the character. Then $\chi^m \circ \lambda_v(t) = t^{\langle m,v\rangle}$. In particular, taking $v = v_\rho$ recovers the same integer that appears in $\mathrm{ord}_{D_\rho}(\chi^m) = \langle m, v_\rho\rangle$.

Theorem 2.16 (Linear equivalence equals homological equivalence for smooth projective toric varieties)**.** *Let X_Σ be a smooth projective toric variety, and let D, D' be divisors on X_Σ. Then the following are equivalent:*

1. *D and D' are linearly equivalent;*
2. *$\mathcal{O}(D) \cong \mathcal{O}(D')$;*

3. $[D] = [D'] \in H_{2n-2}(X_\Sigma, \mathbb{Z})$;
4. $D - D' = (m)$ *for some* $m \in M$.

Proof. We prove the only nontrivial implication (3) ⇒ (1), since the other implications

$$(4) \Leftrightarrow (1) \Rightarrow (2) \Rightarrow (3)$$

hold for divisors on any normal variety.

Assume that $[D] = [D'] \in H_{2n-2}(X_\Sigma, \mathbb{Z})$. By Poincaré duality, this is equivalent to

$$\mathrm{PD}([D]) = \mathrm{PD}([D']) \in H^2(X_\Sigma, \mathbb{Z}).$$

For any divisor E on a smooth variety, the Poincaré dual of its homology class is the first Chern class of the associated line bundle:

$$\mathrm{PD}([E]) = c_1\big(\mathcal{O}(E)\big).$$

Hence

$$c_1\big(\mathcal{O}(D)\big) = c_1\big(\mathcal{O}(D')\big) \in H^2(X_\Sigma, \mathbb{Z}).$$

We now use the following standard fact for smooth projective toric varieties.

Lemma 2.17 (Vanishing and Picard–cohomology isomorphism for toric varieties)**.** *Let X_Σ be a smooth projective toric variety. Then*

$$H^i(X_\Sigma, \mathcal{O}_{X_\Sigma}) = 0 \quad \textit{for all } i > 0,$$

and the first Chern class map induces an isomorphism

$$c_1 : \mathrm{Pic}(X_\Sigma) \xrightarrow{\sim} H^2(X_\Sigma, \mathbb{Z}).$$

Proof. For smooth projective toric varieties, there is a highly nontrivial fact that the Hodge numbers satisfy

$$H^{p,q}(X_\Sigma) = 0 \quad \text{for } p \neq q.$$

In particular,

$$H^i(X_\Sigma, \mathcal{O}_{X_\Sigma}) = H^{0,i}(X_\Sigma) = 0 \quad (i > 0).$$

Applying cohomology to the exponential exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{O}_{X_\Sigma} \xrightarrow{\exp(2\pi i\,\cdot)} \mathcal{O}^*_{X_\Sigma} \longrightarrow 0$$

gives the exact segment

$$H^1(\mathcal{O}_{X_\Sigma}) \longrightarrow \mathrm{Pic}(X_\Sigma) \xrightarrow{c_1} H^2(X_\Sigma, \mathbb{Z}) \longrightarrow H^2(\mathcal{O}_{X_\Sigma}).$$

The vanishing of $H^1(\mathcal{O}_{X_\Sigma})$ and $H^2(\mathcal{O}_{X_\Sigma})$ implies that c_1 is an isomorphism. □

Returning to the theorem, since $c_1 : \mathrm{Pic}(X_\Sigma) \to H^2(X_\Sigma, \mathbb{Z})$ is an isomorphism, the equality

$$c_1\big(\mathcal{O}(D)\big) = c_1\big(\mathcal{O}(D')\big)$$

implies

$$\mathcal{O}(D) \cong \mathcal{O}(D').$$

Therefore D and D' are linearly equivalent, completing the proof. □

Proposition 2.18 (Divisor class exact sequence)**.** *Every divisor on X_Σ is linearly equivalent to a T-invariant divisor. If the rays span $N_\mathbb{R}$, there is an exact sequence*

$$0 \longrightarrow M \longrightarrow \mathbb{Z}^{\Sigma(1)} \longrightarrow \mathrm{Cl}(X_\Sigma) \longrightarrow 0,$$

where $m \mapsto (\langle m, v_\rho\rangle)_\rho$ and $e_\rho \mapsto [D_\rho]$. For a normal variety, $\mathrm{Cl}(X_\Sigma)$ is the group of Weil divisors modulo linear equivalence; it is also denoted $A_{r-1}(X_\Sigma)$.

Remark 2.19. The group $\mathrm{Cl}(X_\Sigma)$ is finitely generated and has the form

$$\mathrm{Cl}(X_\Sigma) \cong \mathbb{Z}^s \oplus H,$$

where H is finite torsion. In particular,

$$\mathrm{Cl}(X_\Sigma)/\mathrm{torsion} \cong \mathbb{Z}^s.$$

Proposition 2.20 (Gauge group from divisor classes)**.** *Applying $\mathrm{Hom}(-, \mathbb{C}^*)$ to the exact sequence above yields*

$$0 \longrightarrow \mathrm{Hom}(\mathrm{Cl}(X_\Sigma), \mathbb{C}^*) \longrightarrow \mathrm{Hom}(\mathbb{Z}^{\Sigma(1)}, \mathbb{C}^*) \longrightarrow \mathrm{Hom}(M, \mathbb{C}^*) \longrightarrow 0.$$

Identifying $\mathrm{Hom}(\mathbb{Z}^{\Sigma(1)}, \mathbb{C}^*) \cong (\mathbb{C}^*)^{\Sigma(1)}$ and $\mathrm{Hom}(M, \mathbb{C}^*) \cong T$, we obtain

$$G \cong \mathrm{Hom}(\mathrm{Cl}(X_\Sigma), \mathbb{C}^*).$$

If $\mathrm{Cl}(X_\Sigma)$ has no torsion, then $G \cong (\mathbb{C}^*)^s$.

Definition 2.21 (Relation lattice)**.** Assume for the moment that X_Σ is smooth and complete. The lattice dual to the divisor class group is

$$\Lambda := \mathrm{Hom}(\mathrm{Cl}(X_\Sigma), \mathbb{Z}) \cong H_2(X_\Sigma, \mathbb{Z}).$$

A chosen integral basis will be denoted $\{C_a\}_{a=1}^s$. In noncomplete examples, the same formulas apply to compact curve classes when the relevant intersection pairing is defined, but the displayed identification should not be used without checking hypotheses.

Theorem 2.22 (Charges as intersection numbers)**.** *Let $Q_{i,a}$ be the GLSM charge matrix associated to X_Σ. Then*

$$Q_{i,a} = D_i \cdot C_a,$$

i.e. the charge of the i-th chiral superfield under the a-th $U(1)$ is equal to the intersection number of the corresponding divisor D_i with the curve class C_a.

This statement uses compatible dual bases of divisor and curve lattices. A change of gauge basis right-multiplies Q by an element of $\mathrm{GL}(s, \mathbb{Z})$ and changes the curve basis by the inverse transformation.

Remark 2.23 (Meaning of the charge–intersection theorem)**.** The content of Definition 2.22 is not to introduce new geometric data, but to identify the GLSM charge matrix with the intersection pairing between divisors and curves.

More precisely, since the relation lattice

$$\Lambda := \mathrm{Hom}\big(\mathrm{Cl}(X_\Sigma), \mathbb{Z}\big) \cong H_2(X_\Sigma, \mathbb{Z})$$

has rank s, one may choose a basis $\{C_a\}_{a=1}^s$ of $H_2(X_\Sigma, \mathbb{Z})$ such that the intersection numbers with the toric divisors satisfy

$$Q_{i,a} = D_i \cdot C_a.$$

Thus the GLSM charge matrix Q is simply the matrix representation of the natural pairing

$$\mathrm{Pic}(X_\Sigma) \times H_2(X_\Sigma, \mathbb{Z}) \longrightarrow \mathbb{Z}$$

in the chosen bases $\{D_i\}$ and $\{C_a\}$.

In particular, the number of $U(1)$ gauge factors in the GLSM equals the rank of $H_2(X_\Sigma, \mathbb{Z})$, and the choice of a basis of H_2 corresponds to a choice of basis for the gauge group.

Proposition 2.24 (Mori cone generators and curve basis)**.** *Let X_Σ be a smooth projective toric variety of dimension r. Then the Mori cone $\overline{\mathrm{NE}}(X_\Sigma) \subset H_2(X_\Sigma, \mathbb{R})$ is generated by the classes of torus-invariant curves*

$$C_\tau := V(\tau), \qquad \tau \in \Sigma(r-1),$$

corresponding to $(r-1)$-dimensional cones of the fan.

Moreover, since

$$\mathrm{rank} H_2(X_\Sigma, \mathbb{Z}) = s = |\Sigma(1)| - r,$$

the invariant curve classes span $H_2(X_\Sigma, \mathbb{R})$. An integral basis of $H_2(X_\Sigma, \mathbb{Z})$ can be computed from their integer relations; it need not be a subset of a chosen generating list without an additional unimodularity check.

Proof. The first statement is a standard and nontrivial result in toric geometry, which we do not prove here: for smooth projective toric varieties, every effective one-cycle is numerically equivalent to a nonnegative linear combination of torus-invariant curves corresponding to $(r-1)$-dimensional cones.

Since these curves generate the Mori cone, their real span equals $H_2(X_\Sigma, \mathbb{R})$. This proves the real spanning statement. Passing from a real basis to an integral basis is an integer linear-algebra problem (for example, via Smith normal form), not a consequence of real independence. □

Remark 2.25 (Intersection coordinates and the charge matrix)**.** Fix any integral basis $\{C_a\}_{a=1}^s$ of $H_2(X_\Sigma, \mathbb{Z})$; its elements may be integer combinations of torus-invariant curves. The intersection pairing induces an identification

$$\mathrm{Cl}(X_\Sigma) \;\cong\; \mathrm{Hom}\big(H_2(X_\Sigma, \mathbb{Z}), \mathbb{Z}\big),$$

and each divisor class D_i is represented by its intersection numbers

$$D_i \;\longleftrightarrow\; \big(D_i \cdot C_1, \ldots, D_i \cdot C_s\big).$$

Under this identification, the GLSM charge matrix Q is precisely the matrix of the intersection pairing between toric divisors and the chosen curve basis:

$$Q_{i,a} = D_i \cdot C_a.$$

Thus the i-th row of Q gives the coordinates of the divisor D_i in the Chow group, while the choice of a basis of $H_2(X_\Sigma, \mathbb{Z})$ corresponds to a choice of basis for the $U(1)^s$ gauge group in the GLSM.

Proposition 2.26 (FI parameters and the effective cone)**.** *Let $Q_{i,a}$ be the GLSM charge matrix and $S = \{v_1, \ldots, v_n\}$ the corresponding set of rays. Let*

$$A(S) := \mathbb{Z}^n / \mathrm{im}(M)$$

be the common divisor-class/Gale-dual lattice (equal to $\mathrm{Cl}(X_\Sigma)$ for a fan using these rays). Then the space of FI parameters is naturally $A(S)_\mathbb{R} \cong \Lambda^*_\mathbb{R}$.

Moreover, the divisor classes of the torus-invariant divisors D_i span a cone

$$A^+(S) := \left\{ \sum_{i=1}^{n} x_i [D_i] \,\middle|\, x_i \geq 0 \right\} \subset A(S)_{\mathbb{R}},$$

called the effective divisor cone. The D-term equations admit a solution if and only if

$$r \in A^+(S).$$

Proof. Set $x_i := |\phi_i|^2 \in \mathbb{R}_{\geq 0}$. Then the D-term equations are equivalent to the linear system

$$Q^T x = r, \qquad x \in \mathbb{R}^n_{\geq 0}.$$

Thus a solution exists if and only if r lies in the image of the positive orthant under the linear map Q^T. By the identification

$$Q_{i,*} \;=\; [D_i] \in A(S),$$

the image of $\mathbb{R}^n_{\geq 0}$ under Q^T is precisely the cone generated by the divisor classes $[D_i]$, namely the effective cone $A^+(S)$. Therefore the D-term equations admit a solution if and only if $r \in A^+(S)$. □

Remark 2.27 (GKZ decomposition and GLSM phases)**.** The effective cone $A^+(S)$ admits a polyhedral decomposition into subcones, called the *GKZ (secondary) decomposition*. Each chamber corresponds to a choice of fan Σ' whose set of rays is S, and hence to a toric variety $X_{\Sigma'}$.

If the FI parameters r lie in the interior of a chamber, the space of supersymmetric ground states is a toric variety $X_{\Sigma'}$. Choosing r in different chambers produces different toric phases of the GLSM, related by wall-crossing transitions when r crosses the boundaries.

Remark 2.28 (Kähler cone and the large volume phase)**.** Among all GKZ chambers, the Kähler cone $\mathcal{K}(X_\Sigma)$ of a fixed toric variety X_Σ selects the chamber corresponding to Σ. If the FI parameters lie in the interior of $\mathcal{K}(X_\Sigma)$, then the space of supersymmetric ground states is precisely X_Σ.

The Kähler cone is the interior of the dual of the Mori cone. If the Mori cone is simplicial and its primitive extremal generators form the chosen integral basis C_a, then the condition

$$r \in \mathcal{K}(X_\Sigma)$$

is simply

$$r_a > 0 \qquad \text{for all } a.$$

In general the Mori cone may have more extremal rays than its dimension, so the large-volume chamber is described by all inequalities $r \cdot C > 0$ for nonzero effective curve classes, not merely by positivity of selected coordinates.

Example 2.29 (The total space of $\mathcal{O}_{\mathbb{CP}^2}(-3)$)**.** We illustrate the charge–intersection correspondence in a non-compact example by reconsidering the total space of the line bundle $\mathcal{O}_{\mathbb{CP}^2}(-3)$.

Let X_Σ be the toric variety associated with the fan of Definition 1.30. The matrices $P|Q$ describing the toric data are

$$P \mid Q = \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ 0 & 0 & 1 & -3 \end{array} \right),$$

where the four rows correspond respectively to the toric divisors

$$D_1 = \{x_1 = 0\}, \quad D_2 = \{x_2 = 0\}, \quad D_3 = \{x_3 = 0\}, \quad D_4 = \{\text{zero section}\}.$$

The unique column of Q represents a curve class

$$C \in H_2(X_\Sigma, \mathbb{Z}),$$

since $\mathrm{rk} H_2(X_\Sigma, \mathbb{Z}) = 1$ in this example. The GLSM charges are therefore $(1, 1, 1, -3)$, in agreement with the GLSM description of $\mathcal{O}_{\mathbb{CP}^2}(-3)$ discussed earlier. Geometrically, the curve C is the zero section over a line in $\mathbb{CP}^2$. It intersects each of the divisors D_1, D_2, D_3 transversely in a single point, hence

$$D_i \cdot C = 1, \qquad i = 1, 2, 3.$$

Since C is contained in the zero section D_4, the intersection number $C \cdot D_4$ is given by the degree of the normal bundle of D_4 restricted to C:

$$C \cdot D_4 = \deg N_{D_4/X_\Sigma}|_C.$$

From the charge matrix we read off

$$C \cdot D_4 = -3,$$

which shows that the normal bundle of the zero section is $\mathcal{O}_{\mathbb{CP}^2}(-3)$. We thus recover the geometric interpretation

$$X_\Sigma \;\cong\; \mathrm{Tot}\big(\mathcal{O}_{\mathbb{CP}^2}(-3)\big),$$

purely from the toric charge–intersection data.

3 Constructions in Toric Geometry

3.1 Orbifolds

Intuition: real independence versus integral independence

A simplicial cone has the correct number of rays, so locally it looks like a finite quotient of affine space. It is smooth only when those primitive rays form an integral basis. The determinant/index measures the gap between these two notions and therefore measures the local finite quotient group.

Definition 3.1 (Finite quotient singularities and orbifolds)**.** Let X be an irreducible variety of dimension n.

(a) A point $p \in X$ is called a *finite quotient singularity* if there exists a finite subgroup $G \subset \mathrm{GL}(n, \mathbb{C})$ such that, in a neighborhood of p, the variety X is analytically isomorphic to the quotient $\mathbb{C}^n/G$ near the origin.

(b) The variety X is called an *orbifold* (or *quasismooth*, or said to have *finite quotient singularities*) if every point $p \in X$ is a finite quotient singularity.

Definition 3.2 (Simplicial cones and simplicial fans)**.** A rational polyhedral cone $\sigma \subset N_{\mathbb{R}}$ is called *simplicial* if it can be generated by linearly independent vectors $v_1, \ldots, v_k$, i.e.

$$\sigma = \mathbb{R}_{\geq 0}\langle v_1, \ldots, v_k \rangle \quad \text{with} \quad \dim \sigma = k.$$

Equivalently, the generators form a basis of the real vector space $\mathrm{Span}_{\mathbb{R}}(\sigma)$.

A fan Σ is called *simplicial* if every cone $\sigma \in \Sigma$ is simplicial.

Proposition 3.3 (Orbifold criterion for toric varieties)**.** *Let X_Σ be a toric variety of dimension r. Then X_Σ is an orbifold if and only if the fan Σ is simplicial.*

Proof. If a full-dimensional cone σ is simplicial, its primitive generators span a finite-index sublattice $N'_\sigma \subset N$. With respect to N'_σ the chart is $\mathbb{A}^r$; changing back to N takes the quotient by the finite character group of N/N'_σ. Hence every simplicial chart has only finite abelian quotient singularities.

Conversely, a finite quotient of a smooth germ is $\mathbb{Q}$-factorial: after pulling back a Weil divisor, a suitable multiple descends from a Cartier divisor (one may average/norm over the finite group). For an affine toric chart U_σ, the local divisor class group at its closed orbit has real rank

$$|\sigma(1)| - \dim \sigma.$$

Thus U_σ is $\mathbb{Q}$-factorial exactly when σ is simplicial. Applying this to every cone proves the converse. □

Remark 3.4 (Local orbifold groups)**.** For a full-dimensional simplicial cone $\sigma = \mathbb{R}_{\geq 0}\langle v_1, \ldots, v_r \rangle$, set

$$H = N/\big(\mathbb{Z}v_1 + \cdots + \mathbb{Z}v_r\big).$$

Then H is finite and the group acting on $\mathbb{C}^r$ is its character group $G = \mathrm{Hom}(H, \mathbb{C}^*)$. For a lower-dimensional cone, replace N by the saturated lattice $N \cap \mathrm{Span}_{\mathbb{R}}(\sigma)$ and retain the complementary torus factor.

Remark 3.5 (Global quotient description of simplicial toric orbifolds)**.** Let Σ be a simplicial fan in $N_{\mathbb{R}}$, and suppose there exists a sublattice

$$N' \subset N$$

such that every top-dimensional cone $\sigma \in \Sigma(r)$ is generated by a $\mathbb{Z}$-basis of N'.

Since $N'_{\mathbb{R}} = N_{\mathbb{R}}$, we may regard Σ as a fan in $N'_{\mathbb{R}}$, which defines a smooth toric variety

$$X_{\Sigma,N'}.$$

However, the acting torus changes from $T = N \otimes \mathbb{C}^*$ to

$$T' = N' \otimes \mathbb{C}^*.$$

The inclusion $N' \hookrightarrow N$ induces a finite morphism of tori

$$T' \longrightarrow T,$$

whose kernel is the finite abelian group N/N'.

Consequently, the original toric variety $X_\Sigma = X_{\Sigma,N}$ is obtained as a global quotient

$$X_\Sigma \;\cong\; X_{\Sigma,N'}/\mathrm{Hom}(N/N', \mathbb{C}^*).$$

This global conclusion requires the stated existence of one common sublattice N' compatible with every maximal cone. Simpliciality alone always gives local finite quotients, but does not by itself supply such a global cover with the same fan.

Example 3.6 (The $\mathbb{Z}_3$ orbifold of $\mathbb{CP}^2$)**.** We consider a concrete orbifold example which will later be used in the construction of the mirror of plane cubic curves.

Recall that the dense torus $T = (\mathbb{C}^*)^2$ embeds into $\mathbb{CP}^2$ by

$$(t_1, t_2) \longmapsto (1, t_1, t_2).$$

Let $\mathbb{Z}_3 \subset (\mathbb{C}^*)^2$ be the subgroup generated by (ω, ω^2), where

$$\omega = e^{2\pi i/3}.$$

This generator extends to an action on $\mathbb{CP}^2$ by coordinatewise multiplication

$$(1, t_1, t_2) \longmapsto (1, \omega t_1, \omega^2 t_2).$$

We denote the quotient by $\mathbb{CP}^2/\mathbb{Z}_3$.

To describe this quotient torically, we must first understand the torus

$$T' := T/\mathbb{Z}_3.$$

The lattice of one-parameter subgroups of T' is strictly larger than that of T. Indeed, the map

$$t \longmapsto (1, t^{1/3}, t^{2/3})$$

defines a one-parameter subgroup of T' which does not lift to T. Hence the lattice

$$N' = N + \mathbb{Z}\big(\tfrac{1}{3}, \tfrac{2}{3}\big)$$

strictly contains N.

Since $N'_{\mathbb{R}} = N_{\mathbb{R}}$, we may take the same fan Σ as for $\mathbb{CP}^2$, but now regarded as a fan in the enlarged lattice N'. The resulting toric variety is denoted

$$X_{\Sigma,N'} = \mathbb{CP}^2/\mathbb{Z}_3.$$

Concretely, the generators of the one-dimensional cones of Σ are

$$(-1,-1), \quad (1,0), \quad (0,1).$$

If we change to coordinates adapted to a basis

$$\left\{\big(\tfrac{2}{3}, \tfrac{1}{3}\big), \big(\tfrac{1}{3}, \tfrac{2}{3}\big)\right\}$$

of N', then the same rays are represented by

$$(2,-1), \quad (-1,2), \quad (-1,-1).$$

To see the orbifold singularities locally, consider the two-dimensional cone

$$\sigma = \operatorname{Cone}\big((2,-1),(-1,2)\big).$$

Since

$$\det\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 3,$$

the vectors $(2,-1)$ and $(-1,2)$ generate a sublattice of index 3 in N'. Therefore the corresponding affine toric variety is

$$X_\sigma \cong \mathbb{C}^2/\mathbb{Z}_3.$$

This explicitly exhibits $\mathbb{CP}^2/\mathbb{Z}_3$ as a toric orbifold with local quotient singularities of type $\mathbb{C}^2/\mathbb{Z}_3$.

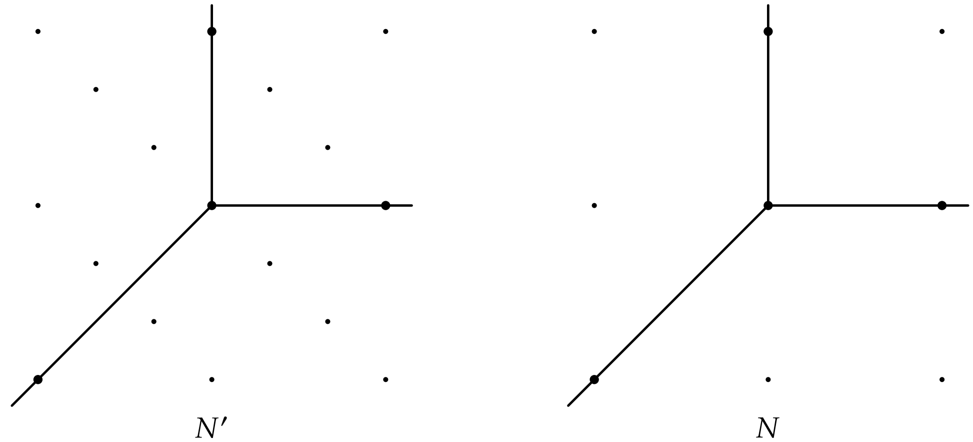

Figure 2: The fans for $\mathbb{CP}^2/\mathbb{Z}_3$ (lattice N') and $\mathbb{CP}^2$ (lattice N).

3.2 Blow-ups and subdivisions

Intuition: a blow-up separates directions of approach

A torus fixed point corresponds to a maximal cone. Inserting a ray through the interior of that cone divides the old set of approach directions into smaller sectors. Geometrically the single fixed point is replaced by a divisor that records those directions. For a smooth cone generated by $v_1, \dots, v_r$, the ordinary point blow-up inserts $v_1 + \cdots + v_r$.

Definition 3.7 (Subdivision of a fan)**.** Let Σ and Σ' be fans in the same lattice N. We say that Σ' *subdivides* Σ if the following conditions hold:

1. $\Sigma(1) \subset \Sigma'(1)$;
2. every cone of Σ' is contained in some cone of Σ.

Proposition 3.8 (Morphisms induced by fan subdivisions)**.** *Let Σ' subdivide Σ and suppose the supports agree: $|\Sigma'| = |\Sigma|$. The identity lattice map $\mathrm{id}_N : N \to N$ maps every cone of Σ' into a cone of Σ, so it induces a proper birational toric morphism*

$$f : X_{\Sigma'} \longrightarrow X_\Sigma.$$

It restricts to the identity on the dense torus.

Proof. The cone-containment condition is exactly the criterion for a lattice map to induce a toric morphism. Equal support gives properness, and the identity on the common dense torus gives birationality. □

Common trap: the map is usually not a coordinate projection

New rays create new Cox coordinates and also change the grading and quotient group. Forgetting the new coordinates is generally not invariant under the new group action. The toric morphism is defined intrinsically by the lattice map; in Cox coordinates it is represented by suitable homogeneous monomials. For the blow-up of $\mathbb{A}^r$ at the origin, the correct map is

$$(x_0, x_1, \dots, x_r) \longmapsto (x_0x_1, \dots, x_0x_r),$$

not $(x_1, \dots, x_r)$.

Proposition 3.9 (Toric blow-up as star subdivision). *Let X_Σ be a smooth projective toric variety of dimension r, and let $p \in X_\Sigma$ be a T-invariant smooth point. Let $\sigma \in \Sigma$ be the r-dimensional cone corresponding to p, with primitive generators*

$$\sigma = \langle v_1, \dots, v_r \rangle \subset N_{\mathbb{R}}.$$

Define a new ray

$$\rho_{r+1} := \mathbb{R}_{\geq 0}\, v_{r+1}, \qquad v_{r+1} := v_1 + \cdots + v_r \in N.$$

Let Σ' be the fan obtained from Σ by replacing the cone σ with the r cones

$$\langle v_1, \dots, \widehat{v_i}, \dots, v_r, v_{r+1} \rangle, \qquad i = 1, \dots, r,$$

and keeping all other cones of Σ unchanged. Then $X_{\Sigma'}$ is the blow-up of X_Σ at the point p.

Proof. The cone σ corresponds to the T-fixed point

$$Z_\sigma \;=\; V(\sigma) \;\subset\; X_\Sigma,$$

which is a smooth point, so σ is generated by a $\mathbb{Z}$-basis of N. The star subdivision replaces the affine toric chart

$$U_\sigma \simeq \mathbb{C}^r$$

by the toric variety associated to $\Sigma'|_\sigma$, which is exactly the toric model of the blow-up of $\mathbb{C}^r$ at the origin.

Gluing with the unchanged cones of Σ yields a birational morphism

$$\pi\colon X_{\Sigma'} \longrightarrow X_\Sigma$$

which is an isomorphism away from Z_σ, and hence realizes the blow-up of X_Σ at the point $p = Z_\sigma$. □

Recipe: blow up a smooth invariant center

Let $\tau = \mathrm{Cone}(v_1, \dots, v_k)$ be a smooth cone. The center is the orbit closure $V(\tau)$, of codimension k. Insert the primitive ray through $v_1 + \cdots + v_k$. Every cone $\sigma \supseteq \tau$ is replaced by the cones obtained by adjoining the new ray and omitting one generator of τ at a time. Cones not containing τ stay unchanged.

Remark 3.10. The following example illustrates concretely the key point behind the construction of the toric morphism: what matters is the existence of a projection that is well defined on the quotient by the group action.

Consider the toric blow-up of $\mathbb{C}^n$ at the origin. In Cox coordinates it can be written as

$$X_{\Sigma'} = (\mathbb{C}^{n+1} \setminus Z)/G, \qquad G = \{(t^{-1}, t, t, \dots, t) \mid t \in \mathbb{C}^*\}.$$

If one attempts to use the naive projection

$$\pi_{\text{naive}} : (x_0, x_1, \dots, x_n) \longmapsto (x_1, \dots, x_n),$$

then this map is not compatible with the G-action. Indeed, for $g = (t^{-1}, t, \dots, t) \in G$ we have

$$\pi_{\text{naive}}(g \cdot (x_0, x_1, \dots, x_n)) = (tx_1, \dots, tx_n) \neq \pi_{\text{naive}}(x_0, x_1, \dots, x_n),$$

so π_{naive} does not descend to a map on the quotient.

To obtain a morphism on the quotient, one must first choose representatives in each G-orbit (i.e. take a slice compatible with the group action). For instance, when $x_0 \neq 0$, every orbit contains a representative of the form

$$(1,\ x_0x_1,\ x_0x_2,\ \dots,\ x_0x_n).$$

On such representatives we define the modified projection

$$\pi(x_0, x_1, \dots, x_n) = (x_0x_1,\ x_0x_2,\ \dots,\ x_0x_n).$$

This map satisfies

$$\pi(g \cdot x) = \pi(x) \qquad (\forall g \in G),$$

and hence is invariant under the G-action, so it induces a well-defined map on the quotient.

In particular, when $x_0 = 0$, the above formula gives $\pi(x) = (0, \dots, 0)$, which corresponds to points on the exceptional divisor mapping to the origin. Thus the apparent paradox that "the origin has no preimage" arises only from using a projection that is not equivariant; once an equivariant representative is chosen, the projection becomes well defined and the exceptional divisor correctly appears as the fiber over the origin.

This shows that, in constructing toric morphisms via Cox coordinates, the essential step is to define the projection on representatives compatible with the group action, rather than using the naive coordinate projection on the ambient affine space.

Remark 3.11 (Exceptional divisor)**.** The new ray ρ_{r+1} corresponds to a torus-invariant divisor

$$E := D_{\rho_{r+1}} \ \subset\ X_{\Sigma'}.$$

This divisor is the *exceptional divisor* of the blow-up. More precisely,

$$E \ =\ \pi^{-1}(Z_\sigma),$$

and $E \simeq \mathbb{P}^{r-1}$, with normal bundle

$$\mathcal{N}_{E/X_{\Sigma'}} \simeq \mathcal{O}_{\mathbb{P}^{r-1}}(-1).$$

The point Z_σ itself is *not* the exceptional locus; rather, it is replaced by the divisor E in the blow-up.

Remark 3.12 (Blow-up of an ideal)**.** A subdivision always gives a proper birational toric morphism when supports agree. It is the normalization of the blow-up of a torus-invariant ideal when the morphism is projective. An arbitrary subdivision need not be projective, so it need not be a blow-up. Star subdivisions associated with smooth invariant centers recover the usual toric blow-ups described above.

Remark 3.13 (Comparison with Algebraic Geometry)**.** In the usual language, the blow-up of a smooth variety X at a point $p \in X$ is defined as

$$\mathrm{Bl}_p X := \operatorname{Proj} \bigoplus_{n \geq 0} \mathfrak{m}_p^n,$$

where $\mathfrak{m}_p$ is the maximal ideal of p. The exceptional divisor is

$$E \simeq \mathbb{P}(T_pX) \simeq \mathbb{P}^{r-1}.$$

The toric construction realizes this abstract definition combinatorially.

Example 3.14 (Resolution of the A_1 singularity)**.** We consider the toric resolution of the orbifold $\mathbb{C}^2/\{\pm 1\}$ at the origin. We choose a lattice N such that the fan of $\mathbb{C}^2/\{\pm 1\}$ consists of the cone

$$\sigma = \mathrm{Cone}(v_1, v_2), \qquad v_1 = (1, 0), \quad v_2 = (1, 2),$$

together with its faces. This cone has determinant 2 and hence corresponds to an A_1 surface singularity.

To resolve this singularity, we insert the ray generated by

$$v_3 = (1, 1),$$

and subdivide σ into two cones

$$\mathrm{Cone}(v_1, v_3) \quad \text{and} \quad \mathrm{Cone}(v_3, v_2).$$

Both cones are basic (their generators form a $\mathbb{Z}$-basis of N), so the resulting fan Σ defines a smooth toric surface X_Σ. This toric variety is the minimal resolution of $\mathbb{C}^2/\{\pm 1\}$ at the singular point.

The new ray $\rho_3 = \mathrm{Cone}(v_3)$ corresponds to a torus-invariant divisor D_3, which is the exceptional divisor of the resolution. The lattice relation

$$v_1 + v_2 - 2v_3 = 0$$

gives the charge vector $(1, 1, -2)$. Equivalently, using the standard toric intersection formula for smooth toric surfaces,

$$v_1 + v_2 = 2v_3 \quad \Longrightarrow \quad D_3^2 = -2.$$

Thus the exceptional divisor is a (-2)-curve, in agreement with the classical resolution of an A_1 singularity.

Remark 3.15. The above computation requires some care in interpretation. Although the Mori cone of X_Σ is one-dimensional, its generators must be represented by *proper* (i.e. compact) curves. Among the torus-invariant divisors D_1, D_2, D_3, only D_3 is compact; the divisors D_1 and D_2 are non-compact and therefore do not represent classes in the Mori cone.

From a cohomological viewpoint, intersection numbers are defined via Poincaré duality, which pairs divisor classes with homology classes represented by compact cycles. Hence, when working on a non-compact toric surface, one is naturally restricted to compact curve classes in H_2. This explains why the Mori cone is generated by the class of D_3 and not by arbitrary combinations of D_1 and D_2.

Consequently, the self-intersection number $D_3^2 = -2$ is intrinsic: it is determined solely by the local toric geometry through the relation $v_1 + v_2 = 2v_3$, and not by an arbitrary choice of a generator of H_2. The other divisors do not play a symmetric role, since they do not correspond to proper curves and therefore do not define intersection classes in the same sense.

Remark 3.16. The previous example extends naturally to the resolution of an A_n singularity. Recall that an A_n surface singularity can be written as the quotient

$$\mathbb{C}^2/\mathbb{Z}_{n+1},$$

where the generator of $\mathbb{Z}_{n+1}$ acts on $\mathbb{C}^2$ by

$$(z_1, z_2) \longmapsto (\omega z_1, \omega^n z_2), \qquad \omega = \exp(2\pi i/(n+1)).$$

A toric model for this singularity is given by the cone

$$\sigma = \mathrm{Cone}((1,0),(1,n+1)) \subset N_{\mathbb{R}}.$$

To obtain a resolution, we subdivide σ by inserting the rays generated by

$$v_i = (1,i), \qquad i = 1,\ldots,n.$$

The resulting fan Σ' defines a smooth toric surface $X_{\Sigma'}$.

For each interior ray v_i with $1 \le i \le n$, the lattice relations

$$v_{i-1} + v_{i+1} - 2v_i = 0$$

imply, as in the A_1 case, that the corresponding torus-invariant divisors satisfy

$$D_i^2 = -2.$$

These divisors are all compact and represent the generators of the Mori cone. Moreover, they intersect according to the pattern

$$D_i \cdot D_{i+1} = 1, \qquad D_i \cdot D_j = 0 \text{ for } |i-j| > 1,$$

so that the divisors $D_1,\ldots,D_n$ form a chain of rational curves. This recovers the classical minimal resolution of an A_n singularity as a chain of n (-2)-curves.

Example 3.17 (A non-simplicial cone and the conifold/node)**.** Let $N \cong \mathbb{Z}^3$ and consider the fan Σ consisting of the cone

$$\sigma = \mathrm{Cone}(v_1,v_2,v_3,v_4) \subset N_{\mathbb{R}}, \qquad v_1 = (1,0,0),\ v_2 = (0,1,0),\ v_3 = (0,0,1),\ v_4 = (1,1,-1),$$

together with all of its faces. Then $X_\Sigma = U_\sigma$ is singular.

Since $\dim\sigma = 3$ but σ is generated by 4 rays, the cone σ is not simplicial. Equivalently, σ cannot be written as $\mathrm{Cone}(w_1,w_2,w_3)$ with w_i linearly independent in $N_{\mathbb{R}}$. Consequently, the singularity of X_Σ is not an orbifold (quotient) singularity.

This singularity is called a *node* in the mathematics literature and a *conifold* singularity in the physics literature. It admits two distinct toric blow-ups (small resolutions) producing smooth toric varieties, obtained by subdividing σ without introducing any new rays. In particular, there is no exceptional divisor. In each subdivision there appears a new two-dimensional cone τ (namely $\tau = \mathrm{Cone}(v_1,v_2)$ in one case and $\tau = \mathrm{Cone}(v_3,v_4)$ in the other), hence an exceptional curve $Z_\tau \subset X_{\Sigma'}$. This curve is a compact one-dimensional toric variety and therefore isomorphic to $\mathbb{CP}^1$. The induced birational map between the two small resolutions is a *flop*.

Theorem 3.18 (Toric resolution of singularities)**.** *For any fan Σ in a lattice N, there exists a refinement $\widetilde{\Sigma}$ such that the induced toric morphism*

$$X_{\widetilde{\Sigma}} \longrightarrow X_\Sigma$$

is a resolution of singularities. That is, $X_{\widetilde{\Sigma}}$ is smooth, and the morphism is proper, birational, and an isomorphism over the smooth locus of X_Σ.

Proof. We only present the idea of the proof. For a cone $\sigma \subset N_{\mathbb{R}}$, the corresponding affine toric variety U_σ is smooth if and only if σ is generated by a part of a $\mathbb{Z}$-basis of N. More generally, U_σ has only orbifold (quotient) singularities if σ is simplicial. Starting from an arbitrary fan Σ, one can subdivide each non-smooth cone by inserting additional rays generated by primitive lattice points. A classical algorithm is to take barycentric subdivisions or stellar subdivisions along carefully chosen lattice vectors. After finitely many steps, every maximal cone becomes basic, producing a smooth fan $\widetilde{\Sigma}$. The morphism $X_{\widetilde{\Sigma}} \to X_\Sigma$ induced by the refinement is proper, birational, and an isomorphism over the smooth locus, hence a resolution of singularities. □

Remark 3.19 (Comments on toric resolution of singularities)**.** Definition 3.18 states that every toric variety admits a toric resolution of singularities. This result is fundamentally combinatorial: singularities of a toric variety are encoded entirely in the structure of its fan, and resolving them amounts to refining the fan so that all cones become smooth (or at least simplicial).

Unlike Hironaka's resolution of singularities for general algebraic varieties, which relies on deep algebro-geometric techniques, the toric case is completely explicit and constructive. The existence of $\widetilde{\Sigma}$ can be viewed as a toric analogue of desingularization by successive blow-ups, but carried out purely at the level of polyhedral geometry.

3.3 Morphisms

Intuition: a toric morphism is a linear map that respects sectors

On the dense tori, a toric morphism is just an integer matrix. It extends over the boundary precisely when every cone in the source lands inside one cone in the target. To test a proposed map, inspect cones rather than trying to guess a formula in homogeneous coordinates.

Definition 3.20. Let Σ be a fan in $N_\mathbb{R}$ and let Σ' be a fan in $N'_\mathbb{R}$. A morphism from Σ to Σ' consists of a homomorphism $\psi : N \to N'$ such that for each $\sigma \in \Sigma$, the image of σ under $\psi \otimes \mathbb{R}$ is contained in some cone of Σ'.

The mapping $\psi : N \to N'$ induces a natural mapping of tori

$$T_N = N \otimes_\mathbb{Z} \mathbb{C}^* \longrightarrow T_{N'} = N' \otimes_\mathbb{Z} \mathbb{C}^*.$$

The cone-containment condition is exactly what ensures that this extends to a map $X_\Sigma \to X_{\Sigma'}$. The global orbifold considered before gives a class of simple examples. In that case, we have N' is a sublattice of N, $\psi : N' \to N$ is the inclusion mapping, $\psi \otimes \mathbb{R}$ is the identity map, and $\Sigma = \Sigma'$.

Example 3.21. Consider the toric variety $X = \mathbb{CP}^1 \times \mathbb{CP}^1$. Its fan $\Sigma \subset \mathbb{R}^2$ has four one-dimensional cones generated by

$$v_1 = (1,0), \qquad v_2 = (0,1), \qquad v_3 = (-1,0), \qquad v_4 = (0,-1).$$

These rays determine four T-invariant divisors and correspondingly four T-invariant curves on X.

There are two natural lattice homomorphisms

$$\pi_1 : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad (x,y) \mapsto x,$$

$$\pi_2 : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad (x,y) \mapsto y,$$

each given by projection onto one of the coordinates.

Both π_1 and π_2 define morphisms of fans $\Sigma \to \Sigma_{\mathbb{CP}^1}$, where $\Sigma_{\mathbb{CP}^1}$ is the standard fan of $\mathbb{CP}^1$ with rays generated by 1 and -1.

Indeed, for every cone $\sigma \in \Sigma$, its image under either projection is contained in one of the cones of $\Sigma_{\mathbb{CP}^1}$. Therefore, each projection induces a morphism of toric varieties

$$X_\Sigma = \mathbb{CP}^1 \times \mathbb{CP}^1 \longrightarrow \mathbb{CP}^1.$$

These are precisely the two coordinate projections onto the first and second factors, respectively.

Example 3.22. Let $n > 0$ and consider the toric surface F_n, whose fan $\Sigma_{F_n} \subset \mathbb{R}^2$ is generated by the rays

$$(1,0), \qquad (0,1), \qquad (-1,n), \qquad (0,-1).$$

Consider the two natural lattice homomorphisms

$$\pi_1 : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad (x,y) \longmapsto x,$$

$$\pi_2 : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad (x,y) \longmapsto y.$$

(1) Projection onto the first factor.

The map π_1 defines a morphism of fans

$$\Sigma_{F_n} \longrightarrow \Sigma_{\mathbb{CP}^1},$$

since for every cone $\sigma \in \Sigma_{F_n}$, the image $(\pi_1 \otimes \mathbb{R})(\sigma)$ is contained in a cone of the standard fan of $\mathbb{CP}^1$. Hence π_1 induces a morphism of toric varieties

$$F_n \longrightarrow \mathbb{CP}^1.$$

This realizes F_n as a $\mathbb{CP}^1$-bundle over $\mathbb{CP}^1$.

(2) Projection onto the second factor.

The map π_2, however, does *not* define a morphism of fans. Indeed, consider the cone in Σ_{F_n} generated by the adjacent rays

$$(-1,n) \quad \text{and} \quad (0,-1).$$

Under π_2, these vectors map to n and -1, whose nonnegative span is all of $\mathbb{R}$. Thus the image of this cone under $\pi_2 \otimes \mathbb{R}$ is the entire line $\mathbb{R}$, which is not contained in any cone of the fan of $\mathbb{CP}^1$. Therefore π_2 is not a morphism of fans, and no toric morphism

$$F_n \longrightarrow \mathbb{CP}^1$$

arises from this projection.

This reflects the fact that F_n is a nontrivial $\mathbb{CP}^1$-bundle over $\mathbb{CP}^1$.

(3) Local triviality.

From the toric viewpoint, the bundle is nevertheless locally trivial. Consider the affine open subset $U \subset \mathbb{CP}^1$ obtained by removing the ray generated by (-1) from the fan of $\mathbb{CP}^1$; this corresponds to $U \simeq \mathbb{C}$.

To restrict the bundle over this open set, we remove the corresponding ray $(-1,n)$ from the fan of F_n. The resulting fan is easily seen to be the product fan of

$$\mathbb{C} \times \mathbb{CP}^1.$$

Performing the same analysis on the other affine chart of $\mathbb{CP}^1$ shows that the bundle is locally a product over both standard affine opens. Hence F_n is a locally trivial $\mathbb{CP}^1$-bundle over $\mathbb{CP}^1$, although it is globally nontrivial.

Example 3.23. We return to the fan Σ of the total space $\mathrm{Tot}(\mathcal{O}_{\mathbb{CP}^2}(-3))$, and observe that it can be constructed directly from the fan Σ' of $\mathbb{CP}^2$.

Each three-dimensional cone of Σ is spanned by

$$(v_1, 1), \qquad (v_2, 1), \qquad (0, 0, 1),$$

where $\{v_1, v_2\}$ is any pair chosen from the set

$$\{(-1,-1),\ (1,0),\ (0,1)\}$$

that spans a two-dimensional cone of Σ'.

Projection onto the first two coordinates defines a lattice homomorphism

$$\pi : \mathbb{Z}^3 \longrightarrow \mathbb{Z}^2, \qquad (x_1, x_2, x_3) \longmapsto (x_1, x_2),$$

which induces a morphism of fans $\Sigma \to \Sigma'$. Consequently, we obtain a toric morphism

$$\pi : \mathrm{Tot}(\mathcal{O}_{\mathbb{CP}^2}(-3)) \longrightarrow \mathbb{CP}^2.$$

This morphism realizes the total space as a locally trivial line bundle over $\mathbb{CP}^2$. The global structure of this bundle, namely that it is $\mathcal{O}_{\mathbb{CP}^2}(-3)$, can be read directly from the toric data.

In general, if the edges of Σ' generated by $v_i \in N$ are lifted to vectors

$$(v_i, k_i) \in N \oplus \mathbb{Z},$$

and an additional ray $(0, \ldots, 0, 1)$ is added, then the resulting toric variety is the total space of the line bundle

$$\mathcal{O}\Big(-\sum k_i D_i\Big),$$

where D_i are the torus-invariant divisors corresponding to the rays v_i.

In the present example, all $k_i = 1$, so the resulting bundle is

$$\mathcal{O}_{\mathbb{CP}^2}(-D_1 - D_2 - D_3) = \mathcal{O}_{\mathbb{CP}^2}(-3).$$

Note that the ray generated by $(0,0,1)$ projects to the zero cone of Σ'. Since the toric subvariety corresponding to the zero cone is the whole base $\mathbb{CP}^2$, we conclude that the divisor $D_{(0,0,1)}$ maps surjectively to $\mathbb{CP}^2$. Indeed, this divisor is precisely the zero section of the line bundle.

The remaining T-invariant divisors project to divisors in $\mathbb{CP}^2$:

$$D_{(-1,-1,1)} = \pi^{-1}(\{x_1 = 0\}), \qquad D_{(1,0,1)} = \pi^{-1}(\{x_2 = 0\}), \qquad D_{(0,1,1)} = \pi^{-1}(\{x_3 = 0\}).$$

Example 3.24. The discussion above can be generalized from line bundles to projective bundles, and even to weighted projective bundles. We now describe an example arising naturally in string theory.

Consider a fan $\Sigma \subset \mathbb{Z}^4$ whose ray generators are the rows of the matrix

$$P = \begin{pmatrix} -1 & -2 & -2 & -3 \\ 1 & 0 & -2 & -3 \\ 0 & 1 & -2 & -3 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -2 & -3 \end{pmatrix}.$$

We leave it as an exercise to determine explicitly the cones of this fan.

Projection onto the first two coordinates defines a lattice map

$$\varphi : \mathbb{Z}^4 \longrightarrow \mathbb{Z}^2, \qquad (x_1, x_2, x_3, x_4) \longmapsto (x_1, x_2),$$

which sends the fan Σ to the fan of the Hirzebruch surface F_2. Hence φ induces a toric morphism

$$X_\Sigma \longrightarrow F_2.$$

The fibers of this morphism are weighted projective spaces $\mathbb{CP}(1, 2, 3)$. Indeed, the rays of Σ that lie in the kernel of φ are

$$(0, 0, 1, 0), \qquad (0, 0, 0, 1), \qquad (0, 0, -2, -3).$$

Restricting to the sublattice $\mathrm{Ker}(\varphi) \cong \mathbb{Z}^2$, these give the rays

$$(1, 0), \qquad (0, 1), \qquad (-2, -3),$$

which are the rays of the fan of the weighted projective space $\mathbb{CP}(1, 2, 3)$. The cones of Σ contained in $\mathrm{Ker}(\varphi)_{\mathbb{R}}$ form the complete fan on these three rays. This kernel fan describes the generic toric fiber, hence the fiber is $\mathbb{CP}(1, 2, 3)$.

This toric variety contains Calabi–Yau hypersurfaces whose fibers over F_2 are elliptic curves embedded in $\mathbb{CP}(1, 2, 3)$. Such constructions provide a standard method for producing elliptic fibrations, and play an important role in F-theory compactifications.

3.4 Local Geometry in Geometric Engineering

Intuition: what toric data is doing in geometric engineering

The fan packages a degeneration and its resolution into integer vectors. A subfan in a kernel describes a fiber; a lattice projection describes the fibration; added rays describe exceptional cycles; and columns of Q record their intersection numbers. This makes shrinking limits and curve volumes computable by linear algebra before one solves any differential equations.

This subsection serves as a very brief introduction to the geometric objects we meet in geometric engineering.

The idea of geometric engineering is to construct geometric models with desired properties so that the resulting string theory, M-theory, or F-theory compactification has the desired physics. Toric geometry provides a useful way to engineer these geometries, as it is easy to perform direct physical computations in this framework.

For example, one way to produce an $\mathcal{N} = 2$ $SU(2)$ gauge theory in four dimensions is to construct a Calabi–Yau threefold X containing a surface F_n, which can be blown down to the base $\mathbb{CP}^1$. We consider type IIA string theory compactified on X. There are two massive states corresponding to D2-branes wrapping the fibers of F_n, with either orientation. In the limit where the fiber shrinks and the base $\mathbb{CP}^1$ becomes large in such a way as to decouple gravity, we obtain a field theory in which these massive states become massless and join with an existing $U(1)$ associated to the volume of the fiber to form an $SU(2)$ vector multiplet.

A local model for this geometry is the canonical bundle of F_n, with F_n embedded as the zero section. This model can be constructed explicitly using toric geometry.

To construct such local Calabi–Yau models, recall Example 7.7.4, where we showed how to build the fan of $\mathcal{O}(K_{\mathbb{CP}^2}) = \mathcal{O}_{\mathbb{CP}^2}(-3)$ from the fan of $\mathbb{CP}^2$, and how this method can be generalized to more general bundles over toric varieties. In particular, we now apply this method to construct the canonical bundle over F_2.

The result can be described in terms of the matrices $P|Q$ for F_2 and for its canonical bundle K_{F_2}:

$$F_2 = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 1 \\ -1 & -2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & -2 \end{array}\right), \qquad K_{F_2} = \left(\begin{array}{ccc|cc} 1 & 0 & 1 & 0 & 1 \\ -1 & -2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & -1 & 1 & 1 & -2 \\ 0 & 0 & 1 & -2 & 0 \end{array}\right).$$

Once this toric data is available, we can directly derive field theory results using mirror symmetry. In particular, the prepotential can be obtained from an appropriate Picard–Fuchs system of differential equations, which can be deduced from the charge matrix Q.

We can also analyze this local geometry directly from the toric data. Projection onto the first coordinate defines a morphism

$$K_{F_2} \longrightarrow \mathbb{CP}^1,$$

for which the divisors D_1 and D_2 are fibers.

To understand the geometry of these divisors, we compute their associated fans as before. Using D_1 as an example, we project the lattice onto

$$\mathbb{Z}^3/\mathbb{Z}\cdot(1,0,1).$$

We use the explicit isomorphism

$$\mathbb{Z}^3/\mathbb{Z}\cdot(1,0,1) \;\cong\; \mathbb{Z}^2, \qquad [(a,b,c)] \longmapsto (c-a,\; b+c-a).$$

Under this identification, the divisor D_1 is seen to be the toric variety associated with a two-dimensional fan whose rays are generated by

$$(1,0), \qquad (1,1), \qquad (1,2).$$

This fan describes the minimal resolution of an A_1 surface singularity. Consequently, the local Calabi–Yau threefold can be viewed as a resolution of an A_1 singularity fibered over $\mathbb{CP}^1$.

More generally, this construction can be extended to produce fibrations of resolutions of A_n singularities over $\mathbb{CP}^1$. The Picard–Fuchs equations for the mirror geometry correspond to the Picard–Fuchs equations governing quantum cohomology, and these relations can be verified directly in many situations.

3.5 Polytopes

We now discuss projective toric varieties and their relationship with polytopes. Our polytopes lie in $M_{\mathbb{R}}$, dual to the space $N_{\mathbb{R}}$ containing fans.

Intuition: a fan and a polytope remember different amounts of data

The normal fan of a polytope remembers the directions of its faces but forgets their positions and lengths. Consequently:

- the fan determines the underlying normal toric variety;
- the polytope determines the variety *together with* a polarization (an ample line bundle and its projective embedding);
- translating a polytope changes only the choice of torus linearization;
- scaling a polytope usually keeps the same variety but replaces L by a tensor power, such as $\mathcal{O}_{\mathbb{P}^2}(1)$ by $\mathcal{O}_{\mathbb{P}^2}(2)$.

So "the toric variety of a polygon" really means the polarized toric surface associated to that polygon.

Recipe: go directly from a lattice polygon to a toric surface

Let $\Delta \subset M_{\mathbb{R}} \simeq \mathbb{R}^2$ be a full-dimensional lattice polygon.

1. Write every edge inequality in primitive form
$$\langle m, v_i \rangle \geq -a_i,$$
where $v_i \in N$ is the primitive *inward* normal.
2. Put the v_i in cyclic order. They are the rays of the normal fan. Adjacent rays span its two-dimensional cones.
3. Check geometry immediately. The fan is complete, and smoothness means $\det(v_i, v_{i+1}) = 1$ for every adjacent pair.
4. Build the surface either by gluing $U_{\mathrm{Cone}(v_i,v_{i+1})}$ or by the Cox recipe in Section 1.4.
5. For the projective embedding, list $m \in \Delta \cap M$ and use all monomials χ^m. In Cox coordinates the monomial for m is
$$x^{D_m} = \prod_i x_i^{\langle m, v_i \rangle + a_i};$$
every exponent is nonnegative by the edge inequalities.

Definition 3.25. An integral polytope in $M_{\mathbb{R}}$ is the convex hull of a finite set of points in M.

In the sequel, we will drop the adjective "integral" and refer to these simply as polytopes.

The r-dimensional polytopes are the data needed to describe projective toric varieties.

Definition 3.26 (Monomial model from a polytope)**.** Let $\Delta \subset M_{\mathbb{R}}$ be an r-dimensional integral polytope. Choose an ordering of the lattice points
$$\Delta \cap M = \{m_0, m_1, \ldots, m_k\}.$$

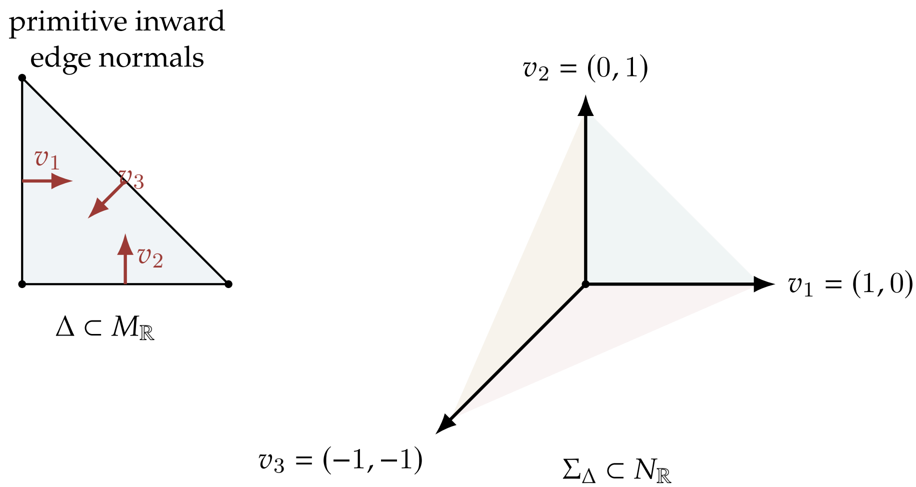

Figure 3: For a polygon, edges become rays of the normal fan and vertices become maximal cones. The standard triangle therefore gives the fan of $\mathbb{P}^2$.

Since $M = \mathrm{Hom}(T, \mathbb{C}^*)$, each m_i can be regarded as a nowhere-vanishing holomorphic function on the torus T. These functions determine a map

$$f : T \longrightarrow \mathbb{CP}^k, \qquad f(t) = (m_0(t), \dots, m_k(t)).$$

We define Y_Δ to be the closure of $f(T)$ in $\mathbb{CP}^k$. With the natural coordinatewise action of T, this is a projective toric variety, possibly nonnormal. This construction is independent of the chosen ordering of $\Delta \cap M$.

Common trap: closure of a monomial map versus the normal toric variety

In arbitrary dimension, Y_Δ need not be normal. Its normalization is the normal toric variety of the normal fan. If Δ is very ample, then Y_Δ is already normal and the two agree. Lattice polygons are normal, so for the two-dimensional polygon computations emphasized here this caveat does not arise.

Remark 3.27. If we write the homogeneous coordinates of $\mathbb{CP}^k$ as $(y_0, \dots, y_k)$, then the image of f is described by the relations among the lattice points m_i. More precisely, if

$$\sum_i a_i m_i = 0 \quad \text{in } M \qquad \text{with} \qquad \sum_i a_i = 0,$$

then the image satisfies the homogeneous equation

$$\prod_{a_i > 0} y_i^{a_i} = \prod_{a_i < 0} y_i^{-a_i}.$$

Thus the monomial model $Y_\Delta \subset \mathbb{CP}^k$ can be defined directly by such binomial equations.

Remark 3.28. Using only a subset $A \subseteq \Delta \cap M$ gives a projective monomial variety Y_A. Its normalization is governed by the convex hull together with the lattice generated by differences $A - A$; omitting lattice points can therefore change normality or even introduce a finite quotient of the dense torus.

Definition 3.29 (Normal fan of a polytope)**.** Assume now that we restrict to normal toric varieties. For each face F of the polytope Δ, define the cone

$$\sigma_F = \{\, v \in N_{\mathbb{R}} \mid \langle m, v\rangle \le \langle m', v\rangle \text{ for all } m \in F,\ m' \in \Delta \,\}.$$

The collection of cones $\{\sigma_F\}$, as F ranges over all faces of Δ, forms a fan in $N_{\mathbb{R}}$, called the *normal fan* of Δ, and is denoted by Σ_Δ.

For a facet F_i written as

$$\langle m, v_i\rangle = -a_i, \qquad \Delta \subseteq \{m : \langle m, v_i\rangle \ge -a_i\},$$

the vector v_i is the primitive inward normal and generates the ray σ_{F_i}. In a polygon, the normal cone at a vertex is simply the cone spanned by the inward normals of the two incident edges.

Theorem 3.30. *Let $\Delta \subset M_{\mathbb{R}}$ be an integral polytope and let Σ_Δ be its normal fan. Then the toric variety X_{Σ_Δ} is the normalization of the monomial model Y_Δ. If Δ is very ample (in particular, for a lattice polygon), then*

$$X_{\Sigma_\Delta} \simeq Y_\Delta.$$

From now on we write

$$\mathbb{CP}_\Delta := X_{\Sigma_\Delta}$$

for the normal toric variety. When Δ is very ample, it is identified with the embedded monomial model $Y_\Delta \subset \mathbb{P}^k$.

Remark 3.31. The isomorphism in the theorem can be described explicitly as follows. Write the facet presentation as

$$\Delta = \{m \in M_{\mathbb{R}} : \langle m, v_i\rangle \ge -a_i,\ i = 1, \dots, n\}.$$

If $(x_1, \dots, x_n)$ are the Cox coordinates on X_{Σ_Δ}, then the map associated to the complete linear system is

$$(x_1, \dots, x_n) \longmapsto \Big(\prod_{i=1}^{n} x_i^{\langle m_0, v_i\rangle + a_i}, \dots, \prod_{i=1}^{n} x_i^{\langle m_k, v_i\rangle + a_i}\Big).$$

The shifts $+a_i$ are essential: they make all exponents nonnegative and make the monomials have the same Cox degree.

Example 3.32 (The standard triangle, completely mechanically)**.** Let

$$\Delta = \mathrm{Conv}\{(0,0), (1,0), (0,1)\}.$$

Its inequalities are

$$x \ge 0, \qquad y \ge 0, \qquad -x - y \ge -1.$$

Thus the primitive inward normals are

$$v_1 = (1,0), \qquad v_2 = (0,1), \qquad v_3 = (-1,-1),$$

with $(a_1, a_2, a_3) = (0,0,1)$. They give the standard fan of $\mathbb{P}^2$. The lattice points $(0,0), (1,0), (0,1)$ give

$$(t_1, t_2) \longmapsto [1 : t_1 : t_2].$$

In Cox coordinates, the same three sections are

$$x_3, \qquad x_1 x_3^0 = x_1, \qquad x_2 x_3^0 = x_2,$$

up to reordering of coordinates. Hence the closure is $\mathbb{P}^2$ with $\mathcal{O}_{\mathbb{P}^2}(1)$.

Example 3.33. We illustrate how projective toric varieties arise from lattice polytopes.

(1) The standard simplex and $\mathbb{CP}^2$.

Let

$$\Delta = \mathrm{Conv}\{(0,0),(1,0),(0,1)\} \subset \mathbb{R}^2.$$

Then

$$\Delta \cap M = \{(0,0),(1,0),(0,1)\}.$$

By the construction of toric varieties from polytopes, the associated map from the torus $T = (\mathbb{C}^*)^2$ is

$$f(t_1,t_2) = (1,t_1,t_2) \in \mathbb{CP}^2.$$

The image of f is dense, and its closure equals the whole projective plane. Hence

$$\mathbb{CP}_\Delta \cong \mathbb{CP}^2.$$

The normal fan Σ_Δ consists of the cones generated by

$$(1,0), \quad (0,1), \quad (-1,-1),$$

which is exactly the standard fan of $\mathbb{CP}^2$. Thus this polytope reproduces the usual toric structure of the projective plane.

(2) A scaled simplex: the Veronese embedding.

Consider instead the scaled triangle

$$\Delta = \mathrm{Conv}\{(0,0),(2,0),(0,2)\}.$$

The shape of the polytope is unchanged, so its normal fan is the same as in part (1). Consequently the associated toric variety is again $\mathbb{CP}^2$.

However, the lattice points are now

$$\Delta \cap M = \{(0,0),(1,0),(2,0),(0,1),(1,1),(0,2)\},$$

six points in total.

Therefore the torus is embedded into $\mathbb{CP}^5$ via

$$(t_1,t_2) \longmapsto (1,t_1,t_1^2,t_2,t_1t_2,t_2^2).$$

This map extends to all of $\mathbb{CP}^2$ as

$$(x_1,x_2,x_3) \longmapsto (x_1^2,x_1x_2,x_2^2,x_1x_3,x_2x_3,x_3^2),$$

which is the classical Veronese embedding of $\mathbb{CP}^2$ of degree 2.

Thus the same toric variety $\mathbb{CP}^2$ can appear with different projective embeddings, depending on the choice of polytope.

(3) Cutting a corner: blow-up of $\mathbb{CP}^2$.

Now consider the quadrilateral

$$\Delta = \mathrm{Conv}\{(0,0),(2,0),(1,1),(0,1)\}.$$

This polytope is obtained from the previous triangle by cutting off the corner at $(0,2)$. Its lattice points are

$$\Delta \cap M = \{(0,0),(1,0),(2,0),(0,1),(1,1)\}.$$

Using these five points, the torus embedding becomes

$$(t_1,t_2) \longmapsto (1,t_1,t_1^2,t_2,t_1t_2) \in \mathbb{CP}^4.$$

Attempting to extend this map to all of $\mathbb{CP}^2$ gives

$$(x_1,x_2,x_3) \longmapsto (x_1^2,x_1x_2,x_2^2,x_1x_3,x_2x_3),$$

but this map is not defined at the point $(0,0,1)$.

To make the map well-defined, we must blow up $\mathbb{CP}^2$ at this point. After performing the blow-up, the map becomes an embedding.

Hence the resulting toric variety is

$$\mathbb{CP}_\Delta \cong F_1,$$

the first Hirzebruch surface, which is the blow-up of $\mathbb{CP}^2$ at one point.

Geometrically, cutting off a corner of the polygon corresponds to inserting a new ray in the fan, which is precisely the toric realization of the blow-up operation.

More generally, blowing up an invariant subvariety corresponds to cutting off the dual face of Δ; for a torus fixed point on a surface, one cuts off a vertex (a corner), not an edge.

We close this section with an easy way to picture the topology of $\mathbb{CP}_\Delta$ directly from Δ. We content ourselves with examples here.

For the first example, consider the map $\mu : \mathbb{CP}^2 \to \mathbb{R}^2$ given by

$$(x_1,x_2,x_3) \longmapsto \left(\frac{|x_2|^2}{|x_1|^2+|x_2|^2+|x_3|^2}, \frac{|x_3|^2}{|x_1|^2+|x_2|^2+|x_3|^2}\right).$$

The image of μ is the set of all $(a,b) \in \mathbb{R}^2$ satisfying $a \geq 0,\ b \geq 0,\ a+b \leq 1$, forming the triangle $\Delta \subset \mathbb{R}^2$. The fiber of a point in the interior of the triangle is a compact torus $S^1 \times S^1$, the fiber over an interior point of an edge is an S^1, and the fiber over a vertex is a point.

An even simpler example is $\mathbb{CP}^1$ with bundle $\mathcal{O}_{\mathbb{CP}^1}(1)$, in which case Δ is the interval $[0,1]$. We have the map $\mu : \mathbb{CP}^1 \to \mathbb{R}$ given by

$$(x_1,x_2) \longmapsto \frac{|x_2|^2}{|x_1|^2+|x_2|^2}.$$

The image of μ is $[0,1]$. The fiber of μ over an interior point of $[0,1]$ is S^1, and the fiber of μ over each of the endpoints is a point. This description leads immediately to the homeomorphism $\mathbb{CP}_\Delta \cong S^2$. The example of $\mathbb{CP}^1$ is actually embedded in the example of $\mathbb{CP}^2$ as the line $x_3 = 0$. Note that the bottom edge of the triangle in the left half of the figure can be identified with $[0,1]$, compatibly with the lattice.

In general, there is a continuous map

$$\mu : \mathbb{CP}_\Delta \longrightarrow \Delta$$

such that the fiber of μ over an interior point of a k–dimensional face of Δ is homeomorphic to $(S^1)^k$.

3.6 Polytopes from Toric Varieties

Quick lookup: the fan–divisor–polytope triangle

Let $\Sigma(1) = \{v_1, \dots, v_n\}$ and let $D = \sum_i a_i D_i$ be a torus-invariant Cartier divisor. Define

$$P_D = \{m \in M_{\mathbb{R}} : \langle m, v_i \rangle \geq -a_i \text{ for all } i\}.$$

Then

$$H^0(X_\Sigma, \mathcal{O}(D)) = \bigoplus_{m \in P_D \cap M} \mathbb{C}\chi^m.$$

If D is ample, P_D is a full-dimensional lattice polytope and its normal fan is exactly Σ. If D is merely nef/basepoint free, its normal fan is a coarsening of Σ. This is the most useful computational bridge between fans and polytopes.

Recipe: construct the polytope attached to a divisor

1. Read the primitive ray generators v_i from the fan.
2. Write the divisor as $D = \sum a_i D_i$.
3. Intersect the half-spaces $\langle m, v_i \rangle \geq -a_i$.
4. Enumerate $P_D \cap M$; these lattice points are a monomial basis of sections.
5. For $m \in P_D \cap M$, the corresponding Cox monomial is $\prod_i x_i^{\langle m, v_i \rangle + a_i}$.

Adding the principal divisor $\mathrm{div}(\chi^u)$ changes a_i by $\langle u, v_i \rangle$ and translates P_D by $-u$; the polarized variety is unchanged.

Now we construct polytopes from projective toric varieties. Suppose we have a toric variety $T \subset X$ embedded in a projective space $\mathbb{CP}^k$. This defines a hyperplane class $\mathcal{O}_X(1)$ on X. We will need to assume that the action of T extends to an action on $\mathbb{CP}^k$, acting by coordinatewise multiplications.

To construct a polytope Δ, we need to choose an isomorphism between $\mathcal{O}_X(1)$ and $\mathcal{O}(D)$, where D is some fixed T–invariant divisor. Making a different choice for D will result in a translation of Δ, so the choice of D is essentially irrelevant.

In the usual way, we identify sections of $\mathcal{O}(D)$ with meromorphic functions f on X such that

$$(f) + D \geq 0,$$

where (f) is the divisor of f. Thus each coordinate function x_i on $\mathbb{CP}^k$ is identified with a meromorphic function f_i on X. The condition that the T–action extends to $\mathbb{CP}^k$ implies that the restriction of f_i to T is a character of T. We let m_i denote this restriction and identify it with an element of M. Then Δ is the convex hull of the $\{m_i\}$.

Remark 3.34. In the construction of a polytope from a projective toric variety, one needs to choose a T-invariant divisor D and an isomorphism

$$\mathcal{O}_X(1) \cong \mathcal{O}(D).$$

This requirement often appears mysterious at first sight, and we briefly explain its necessity.

The goal is to extract from the projective embedding $X \hookrightarrow \mathbb{CP}^k$ a collection of characters of the torus $T \subset X$, since the polytope $\Delta \subset M_{\mathbb{R}}$ is defined as the convex hull of such characters. However, the homogeneous coordinates $x_0, \ldots, x_k$ on $\mathbb{CP}^k$ are not global functions on X; rather, they are global sections of the line bundle $\mathcal{O}_X(1)$. Hence they cannot be directly restricted to T as ordinary functions.

To convert these sections into meromorphic functions on X, we identify the hyperplane bundle with a divisor line bundle via an isomorphism $\mathcal{O}_X(1) \cong \mathcal{O}(D)$. Under this identification, each section x_i corresponds to a section of $\mathcal{O}(D)$, which in turn can be viewed as a meromorphic function f_i satisfying

$$(f_i) + D \geq 0.$$

Restricting f_i to the torus then yields a character $m_i \in M = \mathrm{Hom}(T, \mathbb{C}^*)$. These characters are precisely the data used to define the polytope

$$\Delta = \mathrm{Conv}\{m_0, \ldots, m_k\}.$$

The existence of such a divisor D follows from general properties of toric varieties. For a projective toric variety X, every line bundle arises from a Cartier divisor, and in particular there exists a divisor D with

$$\mathcal{O}_X(1) \cong \mathcal{O}(D).$$

Moreover, one may always choose D to be T-invariant. Different choices of D correspond to linearly equivalent divisors, and hence change the resulting polytope only by a translation in $M_{\mathbb{R}}$. Since translations do not affect the normal fan, the associated toric variety remains unchanged. Therefore the choice of D is essentially irrelevant to the final geometry.

In summary, the divisor D serves as a bridge that allows homogeneous coordinates, which are sections of a line bundle, to be interpreted as meromorphic functions and hence as torus characters. This step is indispensable for translating the projective embedding data of X into the combinatorial data of a polytope.

Remark 3.35. The construction of a polytope depends on the choice of a line bundle, not merely on the toric variety X. More precisely, one should view the input as a pair (X, L), where L is a T-linearized ample line bundle. If X is given together with a projective embedding $X \hookrightarrow \mathbb{CP}^k$, then this embedding determines a natural choice $L = \mathcal{O}_X(1)$. However, one may equally well replace $\mathcal{O}_X(1)$ by any other ample line bundle L' (for instance, $L' = -K_X$ when studying anticanonical hypersurfaces). The resulting polytope is then the polytope $P_{D'}$ associated to $L' \simeq \mathcal{O}_X(D')$.

Example 3.36. We consider $\mathbb{CP}^2$ with hyperplane class identified with $\mathcal{O}(D_1)$. The isomorphism between $\mathcal{O}_X(1)$ and $\mathcal{O}(D_1)$ is defined by division by x_1.

Thus, the coordinates $\{x_1, x_2, x_3\}$ correspond to the meromorphic functions

$$\left(1, \frac{x_2}{x_1}, \frac{x_3}{x_1}\right)$$

respectively. The coordinates on the torus are given by

$$(t_1, t_2) = \left(\frac{x_2}{x_1}, \frac{x_3}{x_1}\right),$$

so the characters are $(0,0)$, $(1,0)$, and $(0,1)$, and we arrive at the polytope Δ that leads us to $\mathbb{CP}^2$.

Example 3.37. For a toric variety X defined by a fan Σ, we have

$$\mathcal{O}(-K_X) \cong \mathcal{O}\left(\sum_{\rho \in \Sigma(1)} D_\rho\right).$$

In particular,

$$\mathcal{O}_{\mathbb{CP}^2}(3) \cong \mathcal{O}(-K_{\mathbb{CP}^2}) \cong \mathcal{O}(D_1 + D_2 + D_3),$$

where D_i is defined by the section x_i of $\mathcal{O}_{\mathbb{CP}^2}(1)$, $i = 1, 2, 3$.

A basis of $\Gamma(\mathcal{O}_{\mathbb{CP}^2}(3))$ is given by the ten homogeneous monomials of degree 3 in x_1, x_2, x_3.

Then with our choice

$$\mathcal{O}_{\mathbb{CP}^2}(3) \cong \mathcal{O}(D_1 + D_2 + D_3),$$

the degree 3 polynomial s is identified with the meromorphic function

$$\frac{s}{x_1 x_2 x_3}$$

on $\mathbb{CP}^2$. The T–action on the vector space

$$V = \left\{ \frac{s}{x_1 x_2 x_3} \,\middle|\, s \in \Gamma(\mathcal{O}_{\mathbb{CP}^2}(3)) \right\}$$

has weights spanning the polytope $\Delta \subset M_{\mathbb{R}}$.

The cones over the proper faces of Δ form its face fan in $M_{\mathbb{R}}$, shown below; this is the normal fan of the polar polytope and gives the toric variety $\mathbb{CP}^2/\mathbb{Z}_3$. We will use this to illustrate mirror symmetry in the next section.

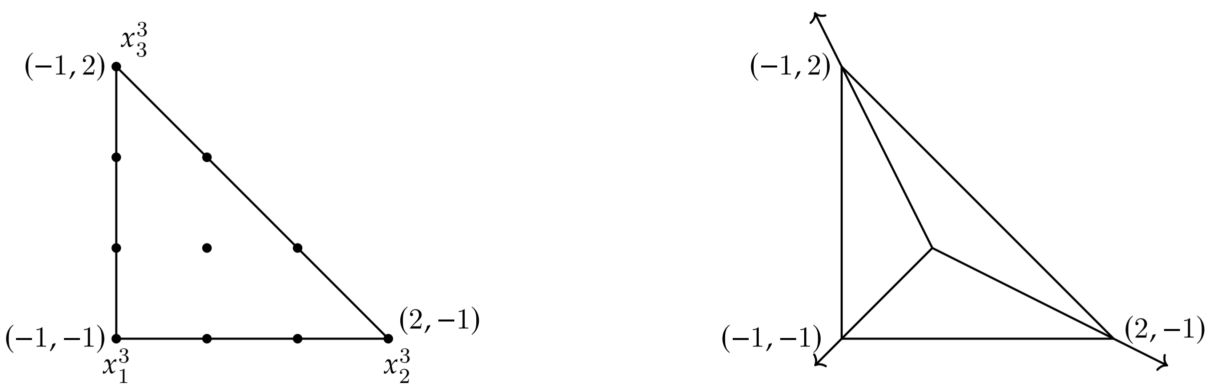

Figure 4: The anticanonical polytope of $\mathbb{CP}^2$ and the fan over its proper faces.

3.7 Mirror Symmetry

Batyrev has introduced a beautiful construction of mirror symmetry for Calabi–Yau hypersurfaces in toric varieties, based on the notion of duality for reflexive polytopes.

Intuition: why polar duality exchanges monomials and divisors

Lattice points of $\Delta \subset M_{\mathbb{R}}$ index monomials in an anticanonical equation. Facets of Δ have primitive normals in N, and those normals index toric divisors of the ambient variety. Passing to the polar polytope turns facet normals into vertices, exchanging these two combinatorial roles. This is the seed of Batyrev's mirror construction.

Definition 3.38. Let P be an n-dimensional polytope. A *facet* of P is a face of dimension $n-1$, i.e. a face of codimension 1.

Definition 3.39. A full dimensional lattice polytope $P \subset M_{\mathbb{R}}$ is called *reflexive* if its facet presentation is of the form

$$P = \{\, m \in M_{\mathbb{R}} \mid \langle m, u_F \rangle \geq -1 \text{ for all facets } F \,\},$$

where each $u_F \in N$ is the primitive inward-pointing normal vector of the facet F.

Proposition 3.40. *Let $P \subset M_{\mathbb{R}}$ be a full dimensional lattice polytope. Assume P is* reflexive. *Then $0 \in M$ is an interior lattice point of P, and it is the unique interior lattice point of P.*

Proof. First, for every facet F we have $\langle 0, u_F \rangle = 0 > -1$, hence $0 \in P^\circ$.

Now let $m \in M \cap P^\circ$ be any interior lattice point. Because m is interior, for every facet inequality we must have $\langle m, u_F \rangle \geq 0$ for all facets F.

Because P is bounded and contains 0 in its interior, its inward facet normals positively span $N_{\mathbb{R}}$: every $v \in N_{\mathbb{R}}$ is a nonnegative linear combination of the u_F. The inequalities $\langle m, u_F \rangle \geq 0$ therefore imply $\langle m, v \rangle \geq 0$ for every $v \in N_{\mathbb{R}}$. Applying this to v and $-v$ gives $\langle m, v \rangle = 0$ for every v, hence $m = 0$. □

Definition 3.41. Let $\Delta \subset M_{\mathbb{R}}$ be a full dimensional lattice polytope containing the origin in its interior. The *polar polytope* (or *dual polytope*) of Δ is the polytope

$$\Delta^\circ = \{\, n \in N_{\mathbb{R}} \mid \langle m, n \rangle \geq -1 \text{ for all } m \in \Delta \,\}.$$

Equivalently, if Δ has facet presentation

$$\Delta = \{\, m \in M_{\mathbb{R}} \mid \langle m, u_F \rangle \geq -1 \text{ for all facets } F \,\},$$

where $u_F \in N$ are the primitive inward-pointing normals to the facets of Δ, then the polar polytope Δ° is the convex hull of these normals:

$$\Delta^\circ = \mathrm{Conv}\{\, u_F \mid F \text{ is a facet of } \Delta \,\} \subset N_{\mathbb{R}}.$$

Definition 3.42. Let X be a normal algebraic variety. We say that X is *Gorenstein* if its canonical divisor K_X is Cartier, or equivalently, if the canonical sheaf ω_X is a line bundle.

Definition 3.43. A projective variety X is called *Fano* if the anticanonical divisor $-K_X$ is ample.

Remark 3.44. For a smooth variety X, the canonical bundle is defined as the line bundle of top degree holomorphic forms,

$$K_X = \Omega_X^n = \wedge^n \Omega_X^1.$$

However, when X has singularities, this definition no longer makes sense globally, since differential forms are only well-defined on the smooth locus.

For a normal variety, the canonical sheaf is defined in general by

$$\omega_X := j_* \Omega^n_{X_{\mathrm{reg}}},$$

where X_{reg} is the smooth part of X and $j : X_{\mathrm{reg}} \hookrightarrow X$ is the inclusion. This extends the usual notion of top degree forms from the smooth locus to the whole variety.

In general, ω_X need not be a line bundle; it is only a reflexive sheaf. The variety X is called *Gorenstein* precisely when this canonical sheaf is invertible, i.e. when it becomes a genuine line bundle (equivalently, when the canonical divisor K_X is Cartier). Thus the Gorenstein condition ensures that the canonical bundle is well-defined even in the presence of singularities.

Remark 3.45. One defining condition for a Calabi–Yau variety is that the canonical line bundle be trivial, namely

$$K_X \cong \mathcal{O}_X.$$

Common definitions also impose smoothness or controlled singularities and cohomological conditions; conventions vary. In any case, every Calabi–Yau variety in the usual sense is Gorenstein, but the converse is false: the Gorenstein condition only requires K_X to be a line bundle, not to be trivial. In particular, a Gorenstein Fano variety is never Calabi–Yau, since for a Fano variety $-K_X$ is ample and therefore K_X cannot be trivial.

Theorem 3.46. *Let Δ be a full dimensional lattice polytope. Then the following statements hold:*

1. *If Δ is reflexive, then X_{Σ_Δ} is Gorenstein Fano and, up to translation, $\Delta = P_{-K_X}$. Conversely, the anticanonical polytope of a Gorenstein toric Fano variety is reflexive. Merely knowing that the underlying variety is Fano is not enough for an arbitrary polarization Δ.*
2. *Δ is reflexive if and only if its polar polytope Δ° is reflexive.*

Batyrev's construction of mirror symmetry can be described as follows. Start with a reflexive polytope $\Delta \subset M_{\mathbb{R}}$. The normal fan Σ_Δ of Δ coincides with the fan obtained by taking the cones over the faces of the polar polytope Δ°.

The anticanonical divisor of the associated toric variety $X_{\Sigma_\Delta} = \mathbb{CP}_\Delta$ is given by

$$-K_{X_{\Sigma_\Delta}} = \sum_{v_i \in \Sigma_\Delta(1)} D_i,$$

where the D_i are the torus invariant prime divisors corresponding to the rays of the fan. Sections of the anticanonical line bundle

$$\mathcal{O}_{X_{\Sigma_\Delta}}\left(\sum_{v_i \in \Sigma_\Delta(1)} D_i\right)$$

define anticanonical hypersurfaces in $\mathbb{CP}_\Delta$. For a generic nondegenerate section, adjunction gives trivial canonical class. When the ambient toric variety is singular, one generally works with a suitable crepant toric partial resolution and the proper transform of the hypersurface. Under the standard hypotheses this produces a Calabi–Yau family, denoted by

$$X \subset \mathbb{CP}_\Delta.$$

By Definition 3.46, the polar polytope Δ° is also reflexive. Therefore the same construction can be applied to Δ°, producing a family of Calabi–Yau hypersurfaces

$$X^\circ \subset \mathbb{CP}_{\Delta^\circ}.$$

The assertion of Batyrev's construction is that these two suitably resolved families form a mirror pair [6].

Remark 3.47. Although a projective toric variety X may be embedded into some projective space using the hyperplane bundle $\mathcal{O}_X(1)$, this embedding is not the data relevant for mirror symmetry. The objects of interest are not the toric varieties themselves, but the Calabi–Yau hypersurfaces

$$Y \in |-K_X|.$$

Mirror symmetry concerns the deformation theory of these hypersurfaces, whose complex moduli are governed by the space of sections $H^0(X, -K_X)$.

The polytope used in Batyrev's construction is designed to encode precisely this linear system. For a line bundle $L = \mathcal{O}_X(D)$ on a toric variety, the associated polytope is determined by the lattice points corresponding to sections of L. Hence, to describe the family of anticanonical hypersurfaces, the appropriate choice is the polytope

$$\Delta = P_{-K_X},$$

whose lattice points parametrize the monomials appearing in sections of $-K_X$.

If one were instead to construct a polytope using $\mathcal{O}_X(1)$, the resulting combinatorial data would encode the embedding of X itself, but would carry no information about the anticanonical linear system and therefore no information about the Calabi–Yau family. Thus the use of the polytope associated to $-K_X$ is essential: it is this polytope that captures the geometry of the hypersurfaces whose mirrors are to be constructed.

Remark 3.48. The reason that a hypersurface defined by a section of $-K_X$ is Calabi–Yau follows from the adjunction formula. Let X be a smooth projective variety and let

$$Y = Z(s) \subset X$$

be the zero locus of a section $s \in H^0(X, L)$ of a line bundle L. Then the canonical bundle of Y is given by

$$K_Y \simeq (K_X + L)|_Y.$$

In the anticanonical case, we take $L = -K_X$. Hence for a hypersurface Y defined by a section of $-K_X$ we obtain

$$K_Y \simeq (K_X + (-K_X))|_Y \simeq \mathcal{O}_Y.$$

Thus Y has trivial canonical bundle. To conclude that Y is a smooth Calabi–Yau variety in the usual sense, one must additionally check smoothness (or choose an appropriate crepant resolution) and the desired cohomological conditions. Generic toric anticanonical hypersurfaces satisfy these conditions in the standard Batyrev setup.

This explains why the anticanonical linear system plays a distinguished role: its hypersurfaces automatically have trivial canonical class by adjunction, subject to the regularity qualifications above.

Theorem 3.49 (Anticanonical equations become hyperplanes)**.** *Let $\Delta = P_{-K_X}$ and set $k = |\Delta \cap M| - 1$. The complete anticanonical linear system defines a toric morphism*

$$\varphi_{|-K_X|} : X \longrightarrow \mathbb{CP}^k.$$

Every anticanonical hypersurface is the pullback of a hyperplane. If $-K_X$ is very ample, $\varphi_{|-K_X|}$ is an embedding; otherwise it may contract strata or normalize its image.

The following standard lemma gives the coordinates of this morphism.

Lemma 3.50. *Let X_Σ be a toric variety, and let*

$$D = \sum a_i D_i$$

be a torus invariant Cartier divisor on X_Σ, where the D_i are the torus invariant prime divisors corresponding to the rays of the fan Σ. Let $\mathcal{O}_{X_\Sigma}(D)$ be the associated line bundle.

Define the polytope

$$P_D = \{\, m \in M_{\mathbb{R}} \mid \langle m, u_i \rangle \geq -a_i \text{ for all } i \,\}.$$

Then the space of global sections of $\mathcal{O}_{X_\Sigma}(D)$ is given by

$$H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D)) = \bigoplus_{m \in P_D \cap M} \mathbb{C} \cdot \chi^m.$$

In particular, a basis of $H^0(X_\Sigma, \mathcal{O}_{X_\Sigma}(D))$ is naturally indexed by the lattice points of the polytope P_D.

Proof of Definition 3.49. Let

$$\Delta \cap M = \{m_0, m_1, \dots, m_k\}$$

be the lattice points of the polytope. The complete linear-system morphism

$$X \longrightarrow \mathbb{CP}^k$$

is obtained by associating to each lattice point m_i the character χ^{m_i}, so that the homogeneous coordinates of $\mathbb{CP}^k$ correspond precisely to these monomials.

A section of the anticanonical bundle is of the form

$$s = \sum_{m \in \Delta \cap M} a_m \, \chi^m.$$

Under the above embedding, the monomials χ^{m_i} become the homogeneous coordinates z_i of $\mathbb{CP}^k$. Hence the section s is represented in $\mathbb{CP}^k$ by the linear expression

$$s = \sum_{i=0}^{k} a_i z_i.$$

Therefore its zero locus is the pullback of the hyperplane $\sum_i a_i z_i = 0$. $\square$

Remark 3.51. The ambient Gorenstein Fano toric variety can be singular. Batyrev's procedure uses a maximal projective crepant partial (MPCP) subdivision of the fan. The resulting ambient space need not be completely smooth in every dimension, but for the usual low-dimensional hypersurface applications a generic proper transform has the required smoothness or mild singularities. "Subdivide the fan" should therefore be read with the three qualifiers *projective*, *crepant*, and *adapted to the hypersurface*.

Example 3.52. We consider the classical example of quintic hypersurfaces in $\mathbb{CP}^4$. The mirror construction of Greene and Plesser describes the mirror family as the family of invariant quintic hypersurfaces in the quotient

$$\mathbb{CP}^4/\mathbb{Z}_5^3,$$

where the effective group is

$$G = \left\{ (\alpha_1, \dots, \alpha_5) \in \mu_5^5 : \prod_{i=1}^{5} \alpha_i = 1 \right\} / \, \mu_5^{\mathrm{diag}} \cong \mathbb{Z}_5^3.$$

The diagonal subgroup acts trivially on projective space, which accounts for the quotient in this formula. This description can be recovered using Batyrev's construction. We start with the fan of $\mathbb{CP}^4$, whose rays are generated by the vectors

$$\begin{pmatrix} -1 & -1 & -1 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Since quintic hypersurfaces in $\mathbb{CP}^4$ are anticanonical, the Batyrev mirror construction applies.

The dual polytope Δ° is the convex hull of the rows of this matrix. The set $\Delta^\circ \cap N$ consists of six lattice points: the five rows above together with the origin, which we denote by v_0. These six lattice points determine an embedding

$$\mathbb{CP}_{\Delta^\circ} \subset \mathbb{CP}^5.$$

Let $(y_0, \dots, y_5)$ be the homogeneous coordinates on $\mathbb{CP}^5$, where y_0 corresponds to the origin v_0. From the relation

$$\sum_{i=1}^{5} v_i = 5v_0,$$

we obtain the defining equation

$$y_1 y_2 y_3 y_4 y_5 = y_0^5$$

for the toric variety $\mathbb{CP}_{\Delta^\circ}$.

Under the change of coordinates

$$y_i = \hat{x}_i^5, \quad i = 1, \dots, 5, \qquad y_0 = \hat{x}_1 \hat{x}_2 \hat{x}_3 \hat{x}_4 \hat{x}_5,$$

this equation is invariant under the automorphism group $\mathbb{Z}_5^3$ described above. Hence this transformation induces an isomorphism

$$\mathbb{CP}_{\Delta^\circ} \cong \mathbb{CP}^4 / \mathbb{Z}_5^3.$$

The anticanonical hypersurfaces in $\mathbb{CP}_{\Delta^\circ}$ are given by linear expressions in the variables y_i. Under the above identification, these correspond precisely to $\mathbb{Z}_5^3$-invariant quintic hypersurfaces in $\mathbb{CP}^4$, as in the Greene–Plesser construction.

Alternatively, the identification $\mathbb{CP}_{\Delta^\circ} \cong \mathbb{CP}^4/\mathbb{Z}_5^3$ can also be deduced by identifying the normal fan of Δ° with the fan obtained from the cones over the proper faces of the polytope Δ corresponding to sections of $\mathcal{O}(5)$ on $\mathbb{CP}^4$, following the general methods of toric geometry.

Master reasoning and reference map

Quick lookup: the construction at a glance

The intrinsic logic of the subject can be recovered from four parallel chains:

$$
\begin{aligned}
(N, M) &\Longrightarrow \sigma \Longrightarrow U_\sigma \Longrightarrow \Sigma \Longrightarrow X_\Sigma, \\
\big(\Sigma(1), \mathrm{Cl}(X_\Sigma)\big) &\Longrightarrow \text{Cox coordinates, irrelevant ideal, and quotient}, \\
D = \sum_i a_i D_i &\Longrightarrow P_D \subset M_{\mathbb{R}} \Longrightarrow H^0(X_\Sigma, \mathcal{O}(D)), \\
\text{reflexive } \Delta &\Longrightarrow \text{anticanonical Calabi–Yau family} \Longleftrightarrow \Delta^\circ \text{ and its mirror family}.
\end{aligned}
$$

Keep the variance straight: fans and one-parameter subgroups live in $N_{\mathbb{R}}$, whereas characters, divisor polytopes, and monomial exponents live in $M_{\mathbb{R}}$. Rays alone do not determine the fan, and a charge matrix alone does not determine the GIT chamber. Those two warnings prevent most reconstruction errors.

When you need	Best next reference
Affine charts, fans, orbit–cone dictionary	Fulton [4], §1–3; Cox–Little–Schenck [2], Chapters 1–3
Cox coordinates, class groups, quotient subtleties	Cox [5]; Cox–Little–Schenck [2], Chapters 4–5
Divisors, line bundles, and polytopes	Cox–Little–Schenck [2], Chapters 4 and 6
Toric surfaces and resolutions	Fulton [4]; Cox–Little–Schenck [2], Chapters 10–11
GLSM, phases, and physics intuition	*Mirror Symmetry* [1], Chapter 7
Secondary fan and wall crossing	Cox–Little–Schenck [2], Chapters 14–15
Reflexive polytopes and mirror hypersurfaces	Batyrev's original paper [6]

Concept Index

References

[1] K. Hori, S. Katz, A. Klemm, R. Pandharipande, R. Thomas, C. Vafa, R. Vakil, and E. Zaslow, *Mirror Symmetry*, Clay Mathematics Monographs, Vol. 1, American Mathematical Society, 2003.

[2] D. Cox, J. Little, and H. Schenck, *Toric Varieties*, Graduate Studies in Mathematics, Vol. 124, American Mathematical Society, 2011.

[3] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics, Vol. 52, Springer-Verlag, 1977.

[4] W. Fulton, *Introduction to Toric Varieties*, Annals of Mathematics Studies, Vol. 131, Princeton University Press, 1993.

[5] D. A. Cox, *The Homogeneous Coordinate Ring of a Toric Variety*, Journal of Algebraic Geometry **4** (1995), 17–50; revised version and erratum, arXiv:alg-geom/9210008.

[6] V. V. Batyrev, *Dual Polyhedra and Mirror Symmetry for Calabi–Yau Hypersurfaces in Toric Varieties*, Journal of Algebraic Geometry **3** (1994), 493–535; arXiv:alg-geom/9310003.